BACK TO BASICS: STRENGTHENING TARGETED DATA POISONING THROUGH SAMPLE SELECTION

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ABSTRACT

Targeted data poisoning attacks manipulate a small subset of training samples to induce a model to misclassify a specific test example as an attacker-chosen class. Existing attacks primarily focus on optimizing perturbations over a predefined set of training samples, which is often selected randomly, while paying limited attention to which samples should be poisoned. In this paper, we show that sample selection can substantially affect attack effectiveness, particularly when the poisoning budget is limited. To understand what makes a training sample effective for poisoning, we incorporate sample selection into the standard bilevel formulation of targeted data poisoning. Our analysis identifies two key characteristics of effective poisoning samples: low model confidence and high representation-space similarity to the target. Motivated by these insights, we propose BASIS (Boundary-Aware Selection Informed by Similarity), a lightweight plug-and-play sample selection method that strengthens existing poisoning attacks without modifying their perturbation optimization. Experiments across multiple attack methods and poisoning budgets demonstrate that BASIS consistently improves attack success rates through sample selection alone, with particularly substantial gains under small poisoning budgets.

1 INTRODUCTION

Building training datasets often requires collecting data from sources that model developers do not fully control, creating opportunities for attackers to introduce carefully modified samples that affect what a model learns. Targeted data poisoning exploits this vulnerability by modifying a small portion of the training data to influence the behavior of the trained model. In targeted data poisoning, the attack is designed to cause a specific, unmodified test example to be classified as an attacker-chosen class (Shafahi et al., 2018; Huang et al., 2020; Geiping et al., 2021) [1]. Poisoning attacks may modify both training samples and their labels, or alter the samples while preserving their original labels. We focus on the latter, known as *clean-label data poisoning*, where the attacker perturbs a limited number of correctly labeled training samples.

Existing data poisoning attacks have largely focused on improving perturbation optimization. Methods such as Bullseye Polytope (Aghakhani et al., 2021), MetaPoison (Huang et al., 2020), Witches' Brew (Geiping et al., 2021), and Sharpness-Aware Attack (He et al., 2024) develop increasingly effective objectives for constructing additive perturbations. In contrast, the training samples used for poisoning are typically chosen arbitrarily, often by taking the first available examples from the attacker-chosen class. This overlooks variation in how individual training examples influence model training (Koh & Liang, 2017). When the poisoning budget is limited, the choice of samples can therefore have a direct impact on attack effectiveness.

In this paper, we integrate sample selection into data poisoning by first defining a general attack objective. We formulate data poisoning as a joint sample selection and perturbation problem, yielding a combinatorial bilevel optimization over a discrete subset of training samples and their perturbations. Existing attacks typically leave sample selection outside the optimization and apply bilevel perturbation optimization to samples chosen randomly in advance. They therefore solve a constrained version of the full problem without searching over potentially more effective samples.

[1] Backdoor attacks embed a trigger into training data so that at test time, triggered inputs are classified as an attacker-chosen label (Gu et al., 2017; Souri et al., 2022). We exclude such attacks from this study.

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Directly solving the combinatorial bilevel problem is computationally intractable. Instead, we use this formulation to derive practical criteria for sample selection. Our analysis identifies two intuitive yet critical properties of effective poisoning samples. Specifically, effective samples should lie near the model's decision boundary while remaining close to the target in the representation space. Moreover, these properties should remain consistent across independently trained models, suggesting that they reflect characteristics of the samples rather than artifacts of a particular model.

Figure 1 illustrates why sample selection matters under a fixed visual perturbation limit. Training samples deep inside the adversarial class (right) require larger perturbations to approach the decision boundary, potentially exhausting the perturbation limit before producing a meaningful boundary shift. Samples closer to the boundary can induce a comparable shift while remaining within the permitted perturbation limits. However, a boundary shift is useful only if it occurs near the target and changes its classification. Effective samples should therefore lie close to both the decision boundary and the target, allowing the perturbation budget to be concentrated where it can most directly move the target into the adversarial class.

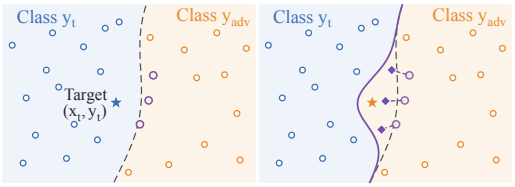


Figure 1: Attack illustration. ○ y_t and ○ y_{adv} denote clean samples; ○···◆ the base-to-poison perturbation; ★ /★ the target pre-/post-attack.

We propose Boundary-Aware Selection Informed by Similarity (BASIS), a lightweight plug-and-play sample selection method that integrates with existing poisoning attacks without modifying their perturbation optimization. Across multiple attacks and budgets, BASIS consistently improves attack success rates. These results show that sample selection can substantially affect attack effectiveness. Finally, we summarize our contributions as follows.

- We extend the standard bilevel formulation of targeted data poisoning to jointly account for sample selection and perturbation optimization, and use this formulation to analyze what makes a training sample effective for poisoning.
- Building on this analysis, we formalize two key characteristics of effective poisoning samples: a small gap between the assigned-class score and the strongest competing-class score, together with high representation-space similarity to the target.
- We introduce BASIS, a lightweight plug-and-play sample selection method that improves attack success over random selection across multiple attacks and poisoning budgets.

2 Related Work

Targeted Clean-label Data Poisoning. Targeted clean-label data poisoning aims to change the prediction of a specific test example by perturbing a small number of correctly labeled training samples. Early methods primarily optimized poisons in representation space. Feature Collision moves poison representations toward the target (Shafahi et al., 2018), while Convex Polytope constructs poisons whose representations surround the target (Zhu et al., 2019). Bullseye Polytope further places the target representation near the center of the poison representations to improve attack scalability (Aghakhani et al., 2021). However, such feature-based attacks become less effective when the victim is trained from scratch (Schwarzschild et al., 2021). Subsequent methods address this issue by integrating victim training into poison optimization. MetaPoison formulates the attack as a bilevel problem and differentiates through multiple training steps (Huang et al., 2020). Witches' Brew instead aligns poison gradients with the target adversarial-loss gradient, avoiding differentiation through the full training process (Geiping et al., 2021). More recently, SAPA uses a sharpness-aware objective to improve attack reliability (He et al., 2024). Together, these methods strengthen targeted clean-label poisoning by improving how perturbations are optimized.

Sample Selection in Adversarial Attacks. Current targeted data poisoning attacks typically use simple sample selection procedures without an explicit criterion for choosing which samples to poison. MetaPoison (Huang et al., 2020) and Bullseye Polytope (Aghakhani et al., 2021), for example, select the first samples from the attacker-chosen class according to the poisoning budget. In triggered backdoor attacks, however, sample selection has received more attention. Criteria based on gradient norms, training instability, feature-space anomalies, and learned poisoning masks have

been used to identify more effective samples for poisoning (Souri et al., 2022; Xia et al., 2022; Gao et al., 2023; Nguyen et al., 2025; Zhu et al., 2026). Together, these works motivate developing more principled sample selection strategies for targeted data poisoning. The closest related approach is Guided Diffusion Poisoning (GDP) (Souri et al., 2024), which generates synthetic samples using diffusion models before perturbation optimization. However, synthetic generation may reduce the visual stealth of the attack, where the goal is to keep poisoned samples visually similar to clean training data. In this work, we instead develop a criterion for selecting effective real samples directly from the existing training set before perturbation optimization.

3 METHOD

3.1 DATA POISONING AS A COMBINATORIAL BILEVEL OPTIMIZATION

Consider a clean training dataset $\mathcal{D}_c = \{(x_i, y_i)\}_{i=1}^n$ consisting of n examples from C classes. Let x_t be a clean, unmodified target example with the ground-truth label y_t. The attacker's objective is to manipulate the victim model's training data such that, after training, the model misclassifies x_t as a chosen adversarial target class y_{adv}, where $y_{\text{adv}} \neq y_t$. The attacker begins by selecting m training examples labeled as y_{adv} to serve as clean selected samples, denoted by $\mathcal{X}_c$. Let Δ denote the collection of perturbations applied to the inputs in $\mathcal{X}_c$. Writing $\mathcal{X}_c = \{(x_j, y_{\text{adv}})\}_{j=1}^m$ and $\Delta = \{\delta_j\}_{j=1}^m$, each perturbation modifies only the input while preserving its label.

To limit the visual impact of the attack, we follow prior work (Geiping et al., 2021; Huang et al., 2020) and constrain every perturbation in Δ by an ℓ_∞-norm budget ε_∞, limiting the change to each pixel value. The resulting poisoning set replaces its selected clean samples[2], yielding

$$\mathcal{X}_p = \{(x_j + \delta_j, y_{\text{adv}})\}_{j=1}^m, \quad \mathcal{D}_p = \left(\mathcal{D}_c \setminus \mathcal{X}_c\right) \cup \mathcal{X}_p.$$

The optimal perturbation depends on the selected samples because each selection induces a different feasible neighborhood under the same perturbation budget. Consequently, different selections can produce substantially different poisoning sets. Thus, the attacker must jointly optimize the combinatorial choice of $\mathcal{X}_c$ and the associated perturbations Δ. Over the admissible sample selections and perturbations described above, the attack can be formulated as the following bilevel objective

$$(\mathcal{X}_c^\star, \Delta^\star) \in \arg\min_{\mathcal{X}_c, \Delta} \quad \mathcal{L}_{\text{adv}}\left(x_t, y_{\text{adv}}; \theta^\star\right) \quad \text{s.t.} \qquad \theta^\star \in \arg\min_\theta \mathcal{L}_{\text{train}}\left(\mathcal{D}_p; \theta\right), \tag{1}$$

where $\mathcal{L}_{\text{adv}}$ denotes the adversarial loss, whose minimization encourages the victim model to predict y_{adv} for x_t, while $\mathcal{L}_{\text{train}}$ denotes the empirical loss used to train the victim model. Optimizing $\mathcal{X}_c$ constitutes the combinatorial component of the outer problem, while optimizing Δ constructs the corresponding poisoned samples. The dependence of the outer objective on the solution of the inner training problem gives the formulation its bilevel structure.

3.2 STRATEGY FOR SELECTING EFFECTIVE SAMPLES

Equation (1) formulates sample selection and perturbation optimization jointly, but using it directly to choose samples would require optimizing the perturbations and retraining the victim for every candidate subset, which is intractable in practice. Our goal is therefore to derive a computationally efficient score for evaluating and selecting clean training samples without solving Equation (1) for each possible subset. Since the selected samples affect the outer adversarial objective through the parameters learned in the inner training problem, we assess each candidate by how a small training update induced by that sample changes the adversarial loss at the target. We consider every clean training sample with label y_{adv} as a candidate and define $\mathcal{X}_c$ as a subset of m selected candidates. For a fixed surrogate model with parameters θ, we compare the parameter gradient induced by candidate x_i with the gradient of the adversarial loss at the target. Specifically, we define

$$g_i = \nabla_\theta \ell\left(x_i, y_{\text{adv}}; \theta\right), \qquad g_t = \nabla_\theta \mathcal{L}_{\text{adv}}\left(x_t, y_{\text{adv}}; \theta\right), \tag{2}$$

where ℓ denotes the per-example cross-entropy loss and $\mathcal{L}_{\text{adv}}(x_t, y_{\text{adv}}; \theta) = \ell(x_t, y_{\text{adv}}; \theta)$. To understand how a candidate affects the attack objective, consider a small parameter update induced by its gradient $\theta' = \theta - \eta g_i$, where $\eta > 0$.

[2]We adopt the replacement setting used in prior clean-label poisoning work (Geiping et al., 2021; Huang et al., 2020), where the attacker modifies existing training examples. This keeps the dataset size and class counts fixed, whereas appending poisons would introduce additional samples and alter the class proportions.

A first-order Taylor expansion of the adversarial loss at x_t around θ gives

$$\mathcal{L}_{\text{adv}}\left(x_t, y_{\text{adv}}; \theta'\right) = \mathcal{L}_{\text{adv}}\left(x_t, y_{\text{adv}}; \theta\right) - \left\langle \nabla_\theta \mathcal{L}_{\text{adv}}\left(x_t, y_{\text{adv}}; \theta\right), \theta - \theta' \right\rangle + \mathcal{O}(\eta^2). \tag{3}$$

The first-order change in adversarial loss is $-\eta\langle g_i, g_t\rangle$, meaning that positive gradient alignment locally decreases the attack objective. Yet if we swap the clean candidate for a perturbed version, the gradient shifts by $\nabla_\theta \ell(x_i + \delta_i, y_{\text{adv}}; \theta) - g_i$. Because δ_i is not known at the time of sample selection, this substitution effect cannot be evaluated directly. Therefore, we treat clean-sample alignment on a fixed surrogate as a selection heuristic prior to optimization, rather than an exact proxy for the impact of replacing a sample with its poisoned counterpart.

To expose the structure of this alignment, we write the model as $z_\theta(x) = W h_\phi(x)$ and $\theta = (W, \phi)$, where $h_\phi(x) \in \mathbb{R}^d$ is the representation before the final linear classifier $W \in \mathbb{R}^{C\times d}$. For brevity, let $h_i = h_\phi(x_i)$ and $h_t = h_\phi(x_t)$. Define

$$r_i = \nabla_{z_\theta(x_i)} \ell\left(x_i, y_{\text{adv}}; \theta\right), \qquad r_t = \nabla_{z_\theta(x_t)} \mathcal{L}_{\text{adv}}\left(x_t, y_{\text{adv}}; \theta\right). \tag{4}$$

The full parameter-gradient alignment can then be written as

$$\langle g_i, g_t\rangle = (r_i^\top r_t)(h_i^\top h_t) + r_i^\top \big(W \frac{\partial h_\phi(x_i)}{\partial \phi} \frac{\partial h_\phi(x_t)}{\partial \phi}^\top W^\top r_t\big), \tag{5}$$

where $\partial h_\phi(x)/\partial\phi$ denotes the derivative of the representation with respect to the parameters ϕ. Further details leading to the above equation are provided in Appendix B.

Evaluating the second term of Equation (5) requires computing parameter gradients for every candidate sample, introducing additional overhead. However, as shown in Section 4.7, the selection criterion derived below from the first term is strongly correlated with this gradient component and yields a candidate set that substantially overlaps with that obtained using exact gradient alignment. We therefore use this criterion without explicitly evaluating the second term.

With this simplification, we examine the first term more closely and identify two characteristics of effective poisoning samples: (i) strong similarity to the target in representation space, quantified by $h_i^\top h_t$ (measured as $R_i = \langle \widehat{h}_i, \widehat{h}_t\rangle$, with $\widehat{h} = h/\|h\|_2$); and (ii) a large $\|r_i\|$ (classifier *residual* magnitude), which signals low model confidence and proximity to the decision boundary. We focus on residual magnitude rather than its alignment with r_t, since most training points have $\|r_i\| \approx 0$ under confident predictions, making the magnitude sufficient for filtering out poor candidates. Moreover, the direction of r_i relative to r_t carries little meaning because the two residuals reflect the model's confidence on different inputs and thus provide no meaningful basis for comparison.

For cross-entropy, $r_i = p_i - e_{y_{\text{adv}}}$, where p_i is the predicted probability vector for x_i and $e_{y_{\text{adv}}}$ is the one-hot vector for y_{adv}; the y_{adv} entry $p_i(y_{\text{adv}}) - 1$ increases in magnitude as confidence drops. We use the competing-class margin

$$M_i = \max_{c \neq y_{\text{adv}}} z_c(x_i) - z_{y_{\text{adv}}}(x_i).$$

A larger M_i indicates a weaker preference for the assigned class relative to its strongest competitor. Furthermore, to reduce model-specific variation, we compute M_i and R_i across K independently trained surrogates. Within each surrogate, both quantities are standardized across candidates and then averaged across surrogates to obtain $\widetilde{M}_i$ and $\widetilde{R}_i$. We then define $s_i = \widetilde{R}_i + \widetilde{M}_i$. This additive form serves as a heuristic rather than a direct approximation of the first term in Equation (5). These steps define BASIS, summarized in Algorithm 1. The m candidates with the largest s_i form $\mathcal{X}_c$, after which their perturbations are optimized using the chosen poisoning method.

Algorithm 1 BASIS

Require: Clean data $\mathcal{D}_c$, target x_t, adversarial label y_{adv}, budget m, K surrogates

Begin
Train K independent surrogates on $\mathcal{D}_c$
for each candidate x_i with $y_i = y_{\text{adv}}$ **do**
  for $k = 1, \ldots, K$ **do**
    $M_i^{(k)} \leftarrow \max_{c\neq y_{\text{adv}}} z_c^{(k)}(x_i) - z_{y_{\text{adv}}}^{(k)}(x_i)$
    $R_i^{(k)} \leftarrow \langle \widehat{h}_i^{(k)}, \widehat{h}_t^{(k)}\rangle$
  end for
end for
Standardize $M_i^{(k)}$ and $R_i^{(k)}$ across candidates within each surrogate, then average across surrogates to obtain $\widetilde{M}_i$ and $\widetilde{R}_i$
$s_i \leftarrow \widetilde{R}_i + \widetilde{M}_i$
Return the m candidates with the largest s_i

4 EXPERIMENTS

We isolate sample selection effects by comparing BASIS with random selection (RAND) on CIFAR-10, paired with Bullseye Polytope (BP) (Aghakhani et al., 2021), Witches' Brew Gradient Matching (GM) (Geiping et al., 2021), and the sharpness-aware poisoning attack (SAPA) (He et al., 2024). Each attack retains its original perturbation objective under identical settings. We assume the attacker has access to the full clean labeled training set $\mathcal{D}_c$. The attacker targets an unmodified test example (x_t, y_t) with an adversarial label $y_{\text{adv}} \neq y_t$. From eligible training points labeled y_{adv}, BASIS selects m examples via Algorithm 1. The poisoning budget is $\rho = m/n$, where n is the dataset size. Following Geiping et al. (2021), we use $K = 20$ surrogates for selection and perturbation optimization while constraining perturbations by $\|\delta_i\|_\infty \leq \varepsilon_\infty = 8/255$. The poisoned samples retain their original labels and replace the corresponding clean data to yield $\mathcal{D}_p$. Our main evaluation trains ConvNet, ResNet20, and VGG13 victims from scratch without data augmentation. We assess ten targets per configuration across six independent training runs and report attack success rate (ASR), the fraction of runs predicting y_{adv} for x_t, alongside clean test accuracy (CTA). Appendix C provides further implementation details.

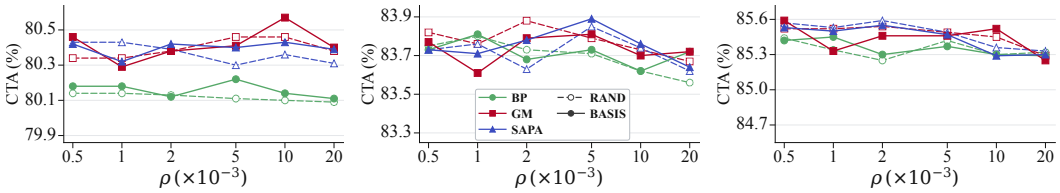
4.1 MAIN RESULTS

Replacing random selection with BASIS more than doubles the mean attack success rate from 20.3% to 44.7% and yields improvements in 103 of 108 configurations in Table 1 across all three victim architectures and both class pairs. Averaging across all 18 configurations in Table 1, BASIS at $\rho = 0.2\%$ achieves a mean ASR of 34.6% compared with 32.5% for RAND at $\rho = 1\%$. This demonstrates a higher overall attack success using 80% fewer modified samples. BASIS delivers the most substantial relative improvements at these smaller budgets where random selection remains ineffective. At larger budgets, the success rates approach saturation in some configurations and leave less room for improvement, although substantial gaps persist in others. Because target sets vary across perturbation objectives, these aggregates represent within-configuration comparisons. Audited confidence intervals in Appendix D confirm the statistical significance of these gains for GM and SAPA, while BP intervals include zero. Since each baseline perturbation algorithm remains unmodified, the results show that BASIS alone substantially strengthens existing data poisoning attacks. Even methods with weaker baseline performance can improve considerably using our BASIS.

The substantial gains in attack success achieved by BASIS come with largely unchanged clean test accuracy. Figure 2 illustrates that BASIS occasionally improves CTA compared to RAND. One possible explanation is our algorithm's preference for weakly classified candidates near or beyond the decision boundary. Perturbing these candidates may change the target prediction with minimal

Table 1: ASR (%, ↑) for data poisoning across poison ratios ρ. We compare RAND and BASIS. Class pairs follow the poison-class→target-class convention ($y_{\text{adv}} \to y_t$).

		dog→bird						**frog→airplane**					
	ρ (%)	**BP**		**GM**		**SAPA**		**BP**		**GM**		**SAPA**	
		RAND	BASIS	RAND	BASIS	RAND	BASIS	RAND	BASIS	RAND	BASIS	RAND	BASIS
ConvNet	0.05	0.0	**13.3**	0.0	**8.3**	0.0	**13.3**	0.0	**5.0**	0.0	0.0	0.0	**3.3**
	0.1	0.0	**28.3**	0.0	**11.7**	0.0	**21.7**	0.0	**5.0**	0.0	**3.3**	0.0	**3.3**
	0.2	1.7	**65.0**	0.0	**28.3**	0.0	**26.7**	0.0	**8.3**	0.0	**3.3**	0.0	**8.3**
	0.5	13.3	**88.3**	1.7	**73.3**	6.7	**48.3**	0.0	**26.7**	0.0	**5.0**	0.0	**5.0**
	1	35.0	**96.7**	21.7	**86.7**	23.3	**81.7**	1.7	**36.7**	3.3	**20.0**	1.7	**21.7**
	2	73.3	**98.3**	43.3	**95.0**	50.0	**100**	10.0	**55.0**	3.3	**35.0**	8.3	**25.0**
ResNet20	0.05	1.7	**15.0**	6.7	**15.0**	6.7	**28.3**	**5.0**	3.3	1.7	**5.0**	0.0	**15.0**
	0.1	11.7	**18.3**	15.0	**35.0**	18.3	**43.3**	**8.3**	5.0	1.7	**23.3**	1.7	**28.3**
	0.2	16.7	**25.0**	31.7	**56.7**	21.7	**53.3**	8.3	**16.7**	0.0	**36.7**	5.0	**20.0**
	0.5	30.0	**46.7**	43.3	**76.7**	55.0	**78.3**	18.3	**25.0**	8.3	**60.0**	3.3	**50.0**
	1	43.3	**55.0**	56.7	**91.7**	65.0	**85.0**	16.7	**40.0**	26.7	**75.0**	21.7	**66.7**
	2	**55.0**	48.3	75.0	**95.0**	95.0	**100**	30.0	**41.7**	45.0	**90.0**	40.0	**83.3**
VGG13	0.05	21.7	**35.0**	0.0	**38.3**	0.0	**28.3**	1.7	**3.3**	0.0	**10.0**	0.0	**18.3**
	0.1	21.7	**28.3**	13.3	**53.3**	11.7	**53.3**	1.7	**8.3**	0.0	**40.0**	0.0	**38.3**
	0.2	21.7	**36.7**	21.7	**56.7**	26.7	**83.3**	1.7	**3.3**	5.0	**46.7**	10.0	**48.3**
	0.5	45.0	**51.7**	55.0	**95.0**	60.0	**95.0**	3.3	**8.3**	16.7	**66.7**	15.0	**66.7**
	1	41.7	**53.3**	76.7	**96.7**	88.3	**100**	13.3	**30.0**	28.3	**83.3**	20.0	**85.0**
	2	61.7	**65.0**	96.7	**100**	100	100	10.0	**16.7**	56.7	**88.3**	53.3	**86.7**



(a) ConvNet (b) ResNet20 (c) VGG13

Figure 2: CTA (%, ↑) under varying poisoning budget ρ for the dog–bird class pair.

impact on overall model behavior. Both selection methods generally exhibit a slight decrease in clean accuracy at larger ρ budgets as the proportion of modified training data increases. Target-level ASR statistics and the complete CTA results corresponding to Figure 2 are provided in Appendix D. Additional evaluations across other CIFAR-10 class pairs and supplementary datasets appear in Appendix E. These extended results support the observed performance trends across the additional evaluated settings. *Overall,* BASIS *substantially enhances the efficacy of existing attacks by improving the choice of poisoned samples while largely preserving clean test accuracy.*

4.2 CROSS-ARCHITECTURE BASE SELECTION AND ENSEMBLE SIZE

We examine the effect of an architecture mismatch between the surrogates used for sample selection (S) and the victim models (V). We vary the selection architecture while matching the perturbation optimization surrogates to the victim architecture [3]. We also test how the number of selection surrogates affects BASIS gains under matched and mismatched settings, fixing perturbation optimization surrogates at $K = 20$. Table 2 reports the results. The matched and mismatched averages use different runs, so comparisons across selection architectures in this table are descriptive. Appendix F.2 further provides a paired comparison with shared targets, crafting settings, and victim-training seeds.

BASIS improves ASR under both matched and mismatched selection architectures. The gains over random selection persist across the evaluated mismatched configurations, showing that useful poison bases can be identified using architectures different from the victim's. At the default ensemble size, matched and mismatched selection achieve similar mean ASR. However, performance varies across architecture pairs. Selection with ConvNet surrogates transfers well to ResNet20 and VGG13 victims and can outperform selection using the victim's own architecture. In the reverse direction, ResNet20 and VGG13 selectors generally perform worse against ConvNet than matched ConvNet selection. These results support the transferability of base selection while showing that matching the selection architecture to the victim does not necessarily yield the most effective bases.

Table 2: ASR (%) of BASIS on dog→bird across architectures and ensemble sizes. Rows are selection architecture S; column groups are victim architecture V. Poison construction uses 20 victim-matched surrogates. Each group contains a shared RAND reference column. Shading marks $S = V$.

Attack	ρ (%)	Selector S	ConvNet						ResNet20						VGG13					
			Rand	$K=1$	$K=3$	$K=10$	$K=20$	$K=30$	Rand	$K=1$	$K=3$	$K=10$	$K=20$	$K=30$	Rand	$K=1$	$K=3$	$K=10$	$K=20$	$K=30$
BP	0.2	ConvNet	1.7	51.7	61.7	60.0	65.0	75.0	16.7	30.0	35.4	36.7	31.2	31.7	21.7	47.9	47.9	47.9	56.2	52.1
		ResNet20		11.7	20.0	23.3	39.6	43.8		16.7	25.0	33.3	31.2	31.7		38.3	38.3	41.7	50.0	51.7
		VGG13		16.7	15.0	23.3	25.0	30.0		25.0	18.3	29.2	35.4	35.4		31.2	35.0	36.7	36.7	38.3
	0.5	ConvNet	13.3	80.0	86.7	86.7	88.3	85.0	30.0	48.3	39.6	43.8	39.6	43.8	45.0	73.3	66.7	73.3	90.0	88.3
		ResNet20		51.7	66.7	65.0	70.0	70.0		29.2	35.4	46.7	46.7	51.7		51.7	60.0	58.3	58.3	54.2
		VGG13		50.0	50.0	58.3	55.0	53.3		33.3	38.3	43.8	37.5	35.4		50.0	47.9	43.8	51.7	53.3
GM	0.2	ConvNet	0.0	23.3	21.7	28.3	28.3	28.3	31.7	53.3	55.0	56.7	55.0	60.0	21.7	81.7	90.0	90.0	90.0	86.7
		ResNet20		1.7	6.7	10.0	15.0	10.0		43.3	50.0	46.7	58.3	63.3		48.3	71.7	76.7	75.0	90.0
		VGG13		1.7	3.3	8.3	10.0	16.7		41.7	43.3	51.7	52.1	52.1		53.3	51.7	70.0	80.0	76.7
	0.5	ConvNet	1.7	23.3	48.3	46.7	50.0	72.9	43.3	66.7	68.3	68.3	75.0	72.9	55.0	98.3	93.3	93.3	100	98.3
		ResNet20		18.3	26.7	33.3	54.2	64.6		75.0	76.7	76.7	76.7	80.0		88.3	96.7	96.7	100	100
		VGG13		20.0	18.3	33.3	66.7	68.8		60.0	71.7	72.9	85.4	81.2		90.0	86.7	95.0	95.0	98.3
SAPA	0.2	ConvNet	0.0	27.1	37.5	50.0	26.7	26.7	21.7	58.3	58.3	56.7	65.0	66.7	26.7	83.3	91.7	95.0	100	98.3
		ResNet20		3.3	15.0	11.7	29.2	31.2		46.7	56.7	55.0	61.7	66.7		56.7	70.0	88.3	85.4	89.6
		VGG13		5.0	8.3	6.7	10.0	16.7		41.7	56.7	61.7	52.1	54.2		53.3	71.7	78.3	83.3	88.3
	0.5	ConvNet	6.7	40.0	46.7	46.7	48.3	48.3	55.0	65.0	68.8	77.1	70.0	83.3	60.0	100	100	96.7	95.0	95.0
		ResNet20		18.3	36.7	46.7	50.0	66.7		70.0	76.7	78.3	78.3	80.0		91.7	100	98.3	100	98.3
		VGG13		23.3	20.0	31.7	30.0	48.3		55.0	68.3	78.3	77.1	79.2		95.0	96.7	96.7	97.9	98.3

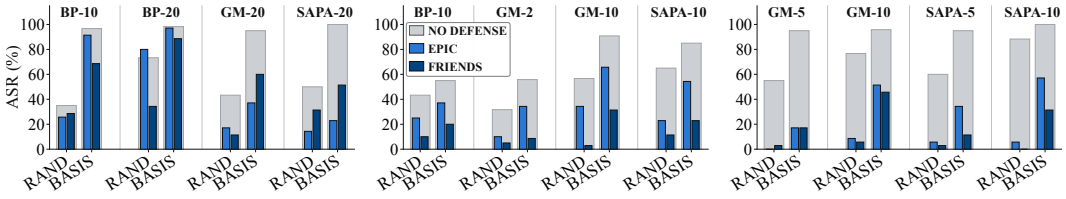
[3] Appendix F.1 considers full architecture mismatch, where selection and perturbation optimization differ from the victim. Appendix F.2 gives paired comparisons of matched vs. mismatched selection architectures.

Larger selection ensembles improve mean ASR and support transfer across architectures. With matched selection, larger K increases mean attack success. Small ensembles can overfit to the specific surrogates used for ranking, whereas averaging scores over independently trained surrogates reduces this sensitivity and generalizes better to unseen victims, potentially aiding cross-architecture transfer under mismatch. Still, gains depend on the configuration, and the largest K adds little under mismatch settings. Overall, ensembling improves ASR on average, but increasing K does not consistently help in every setting.

4.3 Attack Effectiveness under Victim-Side Defenses

We examine whether changing the samples selected for perturbation increases attack success under victim-side defenses. We compare BASIS with RAND, keeping each attack's perturbation objective and optimization fixed, and train victims on the resulting poisoned datasets using EPIC (Yang et al., 2022) or FRIENDS (Liu et al., 2022). These defenses address poisoning through different mechanisms. EPIC drops samples in low-density regions of gradient space, where effective poisons can become isolated during training. Evaluating against EPIC tests whether BASIS gains persist after filtering poisoned samples. FRIENDS optimizes an additive defense perturbation to mitigate the sharp loss regions introduced by poisoned samples. These defenses mitigate poisoning through different mechanisms, enabling us to evaluate the benefits of BASIS under different defense strategies.

Figure 3 shows that BASIS achieves higher ASR than RAND under both defenses in all configurations. FRIENDS generally reduces ASR more than EPIC on ResNet20 and VGG13, while their relative effectiveness varies for ConvNet. Nevertheless, neither defense eliminates the advantage of BASIS. These results show that changing base selection alone can increase attack success even when poisons are not optimized against the defense. Within this non-adaptive setting, evaluations based only on random selection can overstate the protection provided by the tested defenses, so sample selection should be considered when developing and evaluating defenses. Adaptive attacks that optimize selection or perturbations against these defenses remain outside this evaluation. Appendix G.1 provides an additional evaluation with defense strength calibrated to preserve clean accuracy.



(a) ConvNet (b) ResNet20 (c) VGG13

Figure 3: ASR (%, ↑) for RAND and BASIS under EPIC and FRIENDS. Annotations above each group indicate the attack method and poisoning budget in units of 10^{-3}, e.g., BP-20 denotes BP with $\rho = 0.02$. Poisons are crafted without defenses and kept fixed during defended victim training.

4.4 Evaluating the Effect of Victim-Side Data Augmentation

We examine whether the advantage of BASIS persists when victims are trained with different data augmentation methods. The surrogate models used for both base selection and perturbation optimization are trained without data augmentation. Table 3 reports the results.

BASIS maintains its advantage under victim-side data augmentation. Data augmentation generally reduces attack success, but BASIS outperforms RAND in most augmented settings and matches it in the remaining cases. This advantage persists across Crop+Flip, RandAugment, and Cutout, although its magnitude varies across attacks and architectures. These gains persist even though the surrogates used for selection and perturbation optimization are trained without augmentation. Appendix G.2 further evaluates augmentation-aware poison optimization and shows that the selection advantage persists when Crop+Flip is also incorporated during poison optimization.

BASIS preserves the gains in CTA from augmentation. Consistent with Section 4.1, which analyzes the impact of BASIS on CTA relative to RAND when the victim uses no augmentation, we observe the same trend here. Augmentation increases CTA by a similar amount for both selectors, but their ASRs differ substantially, with BASIS achieving higher attack success in most settings. Thus, changing sample selection has little impact on the clean-accuracy gains from augmentation.

Table 3: Victim-side augmentation robustness on dog→bird. Each cell reports RAND/BASIS, with ASR (%) followed by CTA (%) in parentheses. The higher ASR in each pair is bold; ties are unbolded. Budget ρ is expressed as a percentage.

		BP ($\rho = 0.2$)	**BP** ($\rho = 1$)	**BP** ($\rho = 2$)	**GM** ($\rho = 2$)
ConvNet	No aug.	1.7 (80.13) / **65.0** (80.12)	35.0 (80.10) / **96.7** (80.14)	73.3 (80.09) / **98.3** (80.11)	43.3 (80.38) / **95.0** (80.40)
	Crop+Flip	0.0 (85.52) / **5.0** (85.38)	0.0 (85.19) / **20.0** (85.03)	5.0 (85.43) / **30.0** (85.37)	5.0 (86.03) / **45.0** (86.03)
	RandAug.	0.0 (84.19) / 0.0 (84.20)	0.0 (84.27) / **20.0** (83.81)	10.0 (83.80) / **30.0** (83.92)	0.0 (85.66) / **5.0** (85.64)
	Cutout	5.0 (85.43) / 5.0 (85.47)	0.0 (85.42) / **20.0** (85.23)	5.0 (85.58) / **15.0** (85.40)	10.0 (86.12) / **35.0** (85.98)
		GM ($\rho = 0.2$)	**GM** ($\rho = 1$)	**GM** ($\rho = 2$)	**SAPA** ($\rho = 1$)
ResNet20	No aug.	31.7 (83.88) / **56.7** (83.79)	56.7 (83.73) / **91.7** (83.70)	75.0 (83.67) / **95.0** (83.72)	65.0 (83.74) / **85.0** (83.76)
	Crop+Flip	30.0 (89.61) / **60.0** (89.61)	60.0 (89.65) / **65.0** (89.54)	80.0 (89.54) / **95.0** (89.30)	50.0 (89.56) / **80.0** (89.56)
	RandAug.	5.0 (90.39) / **26.3** (90.18)	5.0 (90.36) / **63.2** (90.13)	60.0 (90.07) / **84.2** (89.93)	20.0 (90.31) / **75.0** (90.11)
	Cutout	10.0 (89.99) / **35.0** (89.96)	25.0 (89.91) / **50.0** (89.84)	40.0 (89.91) / **75.0** (89.67)	30.0 (89.88) / **50.0** (89.89)
		BP ($\rho = 0.2$)	**GM** ($\rho = 0.5$)	**GM** ($\rho = 1$)	**SAPA** ($\rho = 1$)
VGG13	No aug.	21.7 (85.25) / **36.7** (85.30)	55.0 (85.49) / **95.0** (85.46)	76.7 (85.45) / **96.7** (85.52)	88.3 (85.36) / **100** (85.29)
	Crop+Flip	10.0 (90.23) / **20.0** (90.16)	25.0 (90.37) / **60.0** (90.60)	60.0 (90.40) / **90.0** (90.58)	50.0 (90.54) / **95.0** (90.47)
	RandAug.	7.1 (91.05) / 7.1 (91.21)	10.0 (91.23) / **50.0** (91.23)	20.0 (91.27) / **65.0** (91.34)	30.0 (91.26) / **80.0** (91.16)
	Cutout	0.0 (90.53) / **5.0** (90.75)	10.0 (90.96) / **25.0** (91.14)	10.0 (91.00) / **70.0** (90.99)	20.0 (91.13) / **85.0** (90.97)

4.5 COMPARISON WITH SAMPLE SELECTION METHODS FOR BACKDOOR ATTACKS

We compare BASIS with sample selection criteria used in backdoor attacks. GRAD picks samples based on their gradient norms (Gao et al., 2023). FORGET ranks candidates by their *forgetting event counts* (Gao et al., 2023), which occur when a sample switches from correct to incorrect classification during training (Toneva et al., 2018). FUS, the Filtering-and-Updating Strategy (Xia et al., 2022), repeatedly trains a model on the current poisoned dataset, records forgetting counts, removes samples with the fewest forgetting events, and replaces them with new candidates. Table 4 reports the comparison of these methods with RAND and BASIS. BASIS achieves the highest ASR in all but one configuration, where it ranks second with a small gap to GRAD. In that configuration, ASR is comparatively high for all selectors, including RAND.

Table 4: ASR (%, ↑) for different sample selection methods. Column values 0.2 and 0.5 denote the poisoning budget ρ in percent (%).

	ConvNet				**ResNet20**			
	GM		**SAPA**		**GM**		**SAPA**	
Method	0.2	0.5	0.2	0.5	0.2	0.5	0.2	0.5
RAND	0.0	1.7	0.0	6.7	31.7	43.3	21.7	55.0
GRAD	10.0	<u>46.7</u>	6.7	<u>48.3</u>	<u>53.3</u>	<u>70.8</u>	45.0	**81.3**
FORGET	<u>15.0</u>	31.7	<u>21.7</u>	40.0	48.3	66.7	<u>46.7</u>	52.1
FUS	8.3	28.3	10.0	20.0	40.0	64.6	40.0	70.8
BASIS	**40.0**	**77.5**	**30.0**	**62.5**	**57.5**	**72.5**	**52.5**	<u>77.5</u>

The performance differences may reflect the distinct attack objectives that these selection criteria were designed to serve. In backdoor attacks, selection methods find samples where adding a trigger makes the model learn a trigger-label association rather than relying on semantic cues (Gao et al., 2023). Data poisoning requires a different approach because it employs no trigger and leaves the test target unchanged. These results show that BASIS is competitive with or exceeds existing selection methods for backdoor attacks by explicitly addressing the unique requirements of data poisoning.

4.6 ABLATION OF THE BASIS RANKING SCORE COMPONENTS

We compare the BASIS score with exact parameter-gradient alignment $\langle g_i, g_t \rangle$ from Equation (5), as well as selection using R or M alone. The R-only variant selects the m candidates with the highest standardized cosine similarity averaged across surrogates, providing a similarity-only baseline under the same scoring procedure as BASIS. Table 5 shows that both margin and similarity are useful individually, but M is considerably more effective than R. The margin-only results therefore show that target-independent candidate difficulty is already a strong selection signal. Exact gradient alignment also performs well, while BASIS achieves higher ASR in some settings. Although exact alignment is the criterion used in our theoretical analysis, it may favor samples that work particularly well on the attacker's surrogate models but transfer less effectively to a separately trained victim. Overall, BASIS frequently achieves the best or second-best ASR, suggesting that its inexpensive score captures much of the useful signal in full parameter-gradient alignment, and combining M and R also improves average ASR over either component alone. Appendix H extends the comparison to

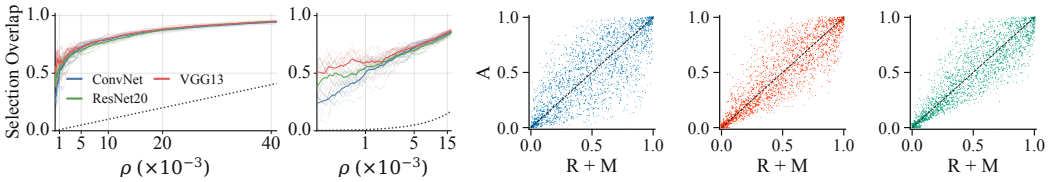
additional selection scores. Results indicate that performance varies with the attack and poisoning budget, while BASIS stays competitive across evaluated settings.

Table 5: Ablation of the selection score across ConvNet, ResNet20, and VGG13. Entries report ASR (%, ↑). The best and second-best results in each column are **bold** and underlined, respectively.

	ConvNet								ResNet20						VGG13			
	BP		GM			SAPA			BP		GM		SAPA		BP		GM	
Method ρ (%)	0.1	0.2	0.1	0.2	0.5	0.1	0.2	0.5	0.1	0.5	0.1	0.2	0.1	0.2	0.1	0.2	0.1	0.2
RAND	0.0	1.7	0.0	0.0	1.7	0.0	0.0	6.7	11.7	30.0	15.0	31.7	18.3	21.7	21.7	21.7	13.3	21.7
R	2.5	22.5	7.5	15.0	20.0	5.0	15.0	25.0	12.5	37.5	37.5	48.0	32.5	42.9	25.0	27.5	55.0	57.5
M	35.0	45.0	17.5	20.8	55.0	15.0	27.5	62.5	17.5	45.0	32.5	55.0	40.0	**55.0**	27.5	25.0	**67.5**	82.5
$\langle g_i, g_t \rangle$	**42.5**	47.5	15.0	35.0	**87.5**	20.0	**40.0**	**82.5**	15.0	37.5	35.0	**57.5**	**47.5**	**55.0**	**35.0**	35.0	55.0	**87.5**
BASIS	32.5	**70.0**	**20.0**	**40.0**	77.5	**27.5**	30.0	62.5	**18.3**	**46.7**	**40.0**	**57.5**	43.3	53.3	28.3	**40.0**	53.3	82.5

4.7 ABLATION ON BASIS AGREEMENT WITH THE EXACT GRADIENT-ALIGNMENT

We examine the empirical agreement between BASIS and $\langle g_i, g_t \rangle$. Also let A_i denote the second term of Equation (5). We first measure the overlap between the candidate sets selected using $s_i = \widetilde{R}_i + \widetilde{M}_i$ and exact parameter-gradient alignment $\langle g_i, g_t \rangle$ across poisoning budgets ρ. We then examine the correlation between s_i and A_i to assess whether higher BASIS scores are also associated with larger values of the omitted term. Figure 4a shows that the overlap between BASIS and exact gradient alignment generally increases with the poisoning budget. By comparison, overlap with RAND remains below 50% even at the largest budget. The correlations between s_i and A_i are 0.749, 0.863, and 0.862 across the three architectures, indicating that candidates ranked highly by BASIS also tend to have larger values of the omitted term. *These results suggest that* BASIS *is a practical alternative to full gradient alignment in the evaluated settings.* Its scoring avoids per-candidate parameter gradients, while total selection cost also includes any required surrogate training and loading (Appendix K).



(a) all (b) ConvNet (c) ResNet20 (d) VGG13

Figure 4: Selection overlap and score correlations for dog→bird poisoning on CIFAR-10. (a) Overlap between subsets selected by BASIS and exact gradient alignment across poisoning budgets ρ, measured as shared candidates divided by m. The dotted line shows overlap with RAND. (b–d) Candidate percentile ranks under s_i and A_i, where A_i is the second term in Equation (5).

Additional Analyses. We further examine the design and practical behavior of BASIS in the appendix. Appendices I and J study sensitivity to score weighting and sample redundancy, while Appendices K and L report computational costs and qualitative selection behavior.

5 CONCLUSION

Sample selection has received comparatively limited attention in targeted clean-label data poisoning. We formulate poisoning as a joint sample-selection and perturbation problem and identify two intuitive yet critical properties of effective poisoning samples: proximity to the model's decision boundary and similarity to the target in representation space. Motivated by this analysis, we introduce BASIS, a lightweight plug-and-play sample-selection method that can be integrated with existing poisoning attacks without modifying their perturbation optimization. Replacing random selection with BASIS substantially improves attack success, particularly at small poisoning budgets, while clean test accuracy remains essentially unchanged. The advantage also persists under architecture mismatch and victim-side defenses. Overall, our results show that sample selection is an important part of targeted data poisoning and should be considered more directly when designing and evaluating poisoning attacks.

Ethics Statement

This work studies targeted data poisoning to better understand weaknesses in machine learning systems and support the evaluation of defenses. We recognize that methods that strengthen poisoning attacks could also be misused to manipulate model behavior. The reported experiments use public benchmark datasets and models trained for research evaluation. Our findings are intended to support authorized security testing and research on more effective defenses.

Reproducibility Statement

Section 3 and Algorithm 1 describe our method, with supporting derivations in Appendix B. Section 4 and Appendix C provide the experimental setup and implementation details. Additional results and statistical analyses are provided in Appendices D and E. Selection-score ablations appear in Table 5, and margin-weight sensitivity is examined in Appendix I. Also, the code used for our experiments will be publicly released.

AI Use Statement

We used generative AI tools to assist with coding, manuscript editing for clarity and readability, methodological and statistical feedback, and interpretation of experimental results. All AI-assisted content was reviewed and verified by the authors who take responsibility for the final manuscript.

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A Additional Related Work

Chen & Koushanfar (2023) survey data and model poisoning attacks and defenses in centralized and federated learning. Several works study clean-label poisoning under threat models related to ours. CLPA (Zhao & Lao, 2022) examines availability attacks that use correctly labeled samples to degrade overall performance, whereas we aim to change the prediction of a specific unmodified target. Blum et al. (2021) theoretically characterize when correctly labeled injected examples can induce errors on particular test points as a function of the hypothesis class, data distribution, and poisoning budget, while Lu et al. (2023) study the reachability of adversarial model behavior under constrained poisoning. Hidden Poison (Di et al., 2023) considers targeted clean-label poisoning in a machine-unlearning setting, where camouflage removal activates the attack after training. Our setting instead assumes standard victim training on a single poisoned dataset without subsequent intervention. Recent advancements in targeted clean-label attacks have also explored first-order efficient approximations for bilevel optimization (Zheng & Li, 2021) and mechanisms that induce dominant features (Zhang et al., 2023). Beyond supervised classification, Foley et al. (2022) adapt gradient alignment to reinforcement learning to induce misbehavior at specific target states by perturbing a small fraction of training observations. Our work complements these attack formulations by focusing on target-conditioned base selection prior to optimization.

Complementary work examines robustness limits and variation in target vulnerability. Hanneke et al. (2022) study optimal learning when an attacker can arbitrarily replace a bounded fraction of training examples. Our setting uses bounded feature perturbations with labels preserved and studies heuristic base selection for deep networks. At the instance level, Xu et al. (2025) relate target poisoning difficulty to clean training dynamics, poison distance, and poisoning budget. Their analysis complements our focus on choosing favorable training bases for a fixed target.

Backdoor attacks train a model to associate a trigger with an attacker-chosen label, allowing the attacker to activate this behavior at test time (Gu et al., 2017). Label-consistent and hidden-trigger attacks show that backdoors can also be constructed using correctly labeled training samples (Saha et al., 2020). Within this setting, the choice of which examples to poison has also been studied. FUS (Xia et al., 2022) uses forgetting behavior to identify effective poisons, while Gao et al. (2023) select hard target-class examples based on loss, gradient norm, and forgetting statistics. Sleeper Agent (Souri et al., 2022) likewise exploits variation among candidate training examples when constructing hidden triggers. Hung-Quang et al. (2023) replace random poisoning with selective poisoning when only limited target-class information is available. Wicked Oddities (Nguyen et al., 2025) extends this by detecting out-of-distribution samples via pretrained representations. Wu et al. (2025) further unify sample selection and trigger construction, making selection depend on trigger and trigger-implanted sample properties, while Zhang et al. (2026) use sample-specific feature alignment for clean-label backdoor construction. Related evidence also appears in text classification, where Kirci & Gursoy (2026) improve clean-label backdoor attacks by poisoning misclassified examples. These works show that poison-set composition affects attack effectiveness, but their criteria aim to reinforce a reusable trigger or target-class behavior. In contrast, our selector is conditioned on the relationship between a candidate clean base and one specific unmodified test target. A sample may be difficult, influential, or helpful for learning a trigger while providing little useful training signal for that specific target.

This distinction is also related to earlier work on example influence and poison importance. Influence functions (Koh & Liang, 2017) quantify how individual training examples affect particular predictions and can be used to construct targeted training perturbations. For base selection, we use target-conditioned quantities before perturbation optimization to determine which clean examples should be passed to a downstream poison optimizer. Yang et al. (2022) exploit variation in poison influence defensively by suppressing particularly harmful examples, while Friendly Noise (Liu et al., 2022) studies how poison signals interact with victim optimization and introduces training-time perturbations to reduce their effect. Poison Forensics (Shan et al., 2022) addresses the retrospective problem of identifying training examples responsible for an observed poisoned prediction. Other defensive work explores feature-map anomaly detection to filter out clean-label poisons (De Gaspari et al., 2024). SAFECLIP (Yang et al., 2024) partitions image-caption pairs into safe and risky sets using cross-modal representation similarity during CLIP pre-training. Its similarity criterion assesses image-caption consistency for defense, whereas ours measures candidate-target similarity for base selection. Guided Diffusion Poisoning (GDP) (Souri et al., 2024) is closest to our focus

on initialization as it synthesizes favorable poison bases with a diffusion model before downstream poison optimization. Our setting instead selects favorable bases from the existing clean training set without generating new examples.

Overall, prior work shows that poisoning effectiveness depends on which training examples are modified, their geometry and influence, and in backdoor settings, their interaction with the trigger. Our formulation isolates the complementary problem of selecting existing adversarial-class examples for a particular target before any poison perturbations are optimized. The selected bases are then passed to the downstream attack without changing its objective, allowing base quality and poison construction to be studied separately.

B Gradient Alignment for Base-Sample Selection

We provide a detailed derivation of the gradient-alignment expression used in Section 3. Consider a clean candidate sample x_i with label y_{adv} and the target sample x_t. At a fixed model state θ, define their parameter gradients as

$$g_i = \nabla_\theta \ell\big(x_i, y_{\text{adv}}; \theta\big), \qquad g_t = \nabla_\theta \mathcal{L}_{\text{adv}}\big(x_t, y_{\text{adv}}; \theta\big).$$

Both losses are evaluated with respect to the adversarial class y_{adv}, since the attack seeks to make x_t be predicted as this class rather than its ground-truth class y_t. We first show how the candidate gradient affects the adversarial loss at the target sample. Consider a sufficiently small gradient-descent update induced by x_i, given by $\theta' = \theta - \eta g_i$, where $\eta > 0$. A first-order Taylor expansion of the adversarial loss around θ gives

$$\mathcal{L}_{\text{adv}}\big(x_t, y_{\text{adv}}; \theta'\big) = \mathcal{L}_{\text{adv}}\big(x_t, y_{\text{adv}}; \theta\big) + \big\langle \nabla_\theta \mathcal{L}_{\text{adv}}\big(x_t, y_{\text{adv}}; \theta\big), \theta' - \theta \big\rangle + \mathcal{O}\big(\|\theta' - \theta\|_2^2\big).$$

Substituting $\theta' - \theta = -\eta g_i$ and using the definition of g_t, the first-order term becomes

$$\big\langle \nabla_\theta \mathcal{L}_{\text{adv}}\big(x_t, y_{\text{adv}}; \theta\big), \theta' - \theta \big\rangle = \big\langle g_t, -\eta g_i \big\rangle = -\eta \langle g_t, g_i \rangle.$$

Moreover, $\|\theta' - \theta\|_2^2 = \eta^2 \|g_i\|_2^2$, and therefore the remainder is $\mathcal{O}(\eta^2)$. Hence,

$$\mathcal{L}_{\text{adv}}\big(x_t, y_{\text{adv}}; \theta'\big) = \mathcal{L}_{\text{adv}}\big(x_t, y_{\text{adv}}; \theta\big) - \eta \langle g_i, g_t \rangle + \mathcal{O}(\eta^2).$$

Since $\eta > 0$, a positive $\langle g_i, g_t \rangle$ produces a negative first-order change in the adversarial loss. A larger positive inner product therefore indicates that the update induced by x_i is more strongly aligned with decreasing the attack objective at x_t.

We next expose the structure of this inner product. We separate the representation network from the final linear classifier and write $z_\theta(x) = W h_\phi(x)$ and $\theta = (W, \phi)$, where $h_\phi(x) \in \mathbb{R}^d$ is the representation before the classifier and $W \in \mathbb{R}^{C \times d}$ contains the classifier weights. A bias term, if present, can be absorbed by appending a constant coordinate to $h_\phi(x)$ and the corresponding bias column to W; d then denotes the augmented feature dimension. We use $h_i = h_\phi(x_i)$ and $h_t = h_\phi(x_t)$. Define the gradients of the candidate and adversarial losses with respect to their logits as

$$r_i = \nabla_{z_\theta(x_i)} \ell\big(x_i, y_{\text{adv}}; \theta\big), \qquad r_t = \nabla_{z_\theta(x_t)} \mathcal{L}_{\text{adv}}\big(x_t, y_{\text{adv}}; \theta\big).$$

For softmax cross-entropy, let $p_\theta(x) = \mathrm{softmax}(z_\theta(x))$. For an arbitrary input x evaluated against y_{adv}, the loss can be written as

$$\ell = -\log p_{\theta, y_{\text{adv}}}(x) = -z_{\theta, y_{\text{adv}}}(x) + \log\big(\sum_{c=1}^{C} \exp(z_{\theta,c}(x))\big).$$

Differentiating with respect to a logit $z_{\theta,c}(x)$ gives

$$\frac{\partial \ell}{\partial z_{\theta,c}(x)} = -\mathbf{1}\big[c = y_{\text{adv}}\big] + \frac{\exp(z_{\theta,c}(x))}{\sum_{k=1}^{C} \exp(z_{\theta,k}(x))} = p_{\theta,c}(x) - \mathbf{1}\big[c = y_{\text{adv}}\big].$$

Collecting the derivatives over all C logits yields $\nabla_z \ell = p_\theta(x) - e_{\text{adv}}$, where e_{adv} is the one-hot vector corresponding to y_{adv}. Thus, under softmax cross-entropy, $r_i = p_\theta(x_i) - e_{\text{adv}}$ and $r_t = p_\theta(x_t) - e_{\text{adv}}$. We now propagate these logit gradients through the final classifier. Let $w_c^\top$

denote row c of W. Since $z_{\theta,c}(x) = w_c^\top h_\phi(x) = \sum_{j=1}^d w_{c,j} h_j$, we have $\partial z_{\theta,c}/\partial w_{c,j} = h_j$, while $\partial z_{\theta,c'}/\partial w_{c,j} = 0$ for $c' \neq c$. Therefore,

$$\frac{\partial \ell}{\partial w_{c,j}} = \sum_{c'=1}^{C} \frac{\partial \ell}{\partial z_{\theta,c'}} \frac{\partial z_{\theta,c'}}{\partial w_{c,j}} = \frac{\partial \ell}{\partial z_{\theta,c}} h_j = r_c h_j.$$

Writing $r = \nabla_z \ell$ and $h = h_\phi(x)$, collecting all coordinates gives the classifier-weight gradient

$$\nabla_W \ell = r h^\top.$$

Applying this result to the candidate and target gives

$$\nabla_W \ell\big(x_i, y_{\text{adv}}; \theta\big) = r_i h_i^\top, \qquad \nabla_W \mathcal{L}_{\text{adv}}\big(x_t, y_{\text{adv}}; \theta\big) = r_t h_t^\top.$$

We next obtain the gradients with respect to the representation parameters ϕ. Since $z = Wh$, for each representation coordinate h_j,

$$\frac{\partial \ell}{\partial h_j} = \sum_{c=1}^{C} \frac{\partial \ell}{\partial z_c} \frac{\partial z_c}{\partial h_j} = \sum_{c=1}^{C} r_c W_{c,j},$$

which in vector form gives $\nabla_h \ell = W^\top r$. Because $h_\phi(x)$ depends on ϕ, applying the chain rule once more gives

$$\nabla_\phi \ell = \big(\frac{\partial h_\phi(x)}{\partial \phi}\big)^\top \nabla_h \ell = \big(\frac{\partial h_\phi(x)}{\partial \phi}\big)^\top W^\top r.$$

Therefore, for the candidate and target,

$$\nabla_\phi \ell\big(x_i, y_{\text{adv}}; \theta\big) = \left(\frac{\partial h_\phi(x_i)}{\partial \phi}\right)^\top W^\top r_i, \qquad \nabla_\phi \mathcal{L}_{\text{adv}}\big(x_t, y_{\text{adv}}; \theta\big) = \left(\frac{\partial h_\phi(x_t)}{\partial \phi}\right)^\top W^\top r_t.$$

Since $\theta = (W, \phi)$, the full parameter gradient consists of the classifier-weight and representation-parameter components. Consequently, their inner product can be separated as

$$\langle g_i, g_t \rangle = \big\langle \operatorname{vec}(r_i h_i^\top), \operatorname{vec}(r_t h_t^\top) \big\rangle + \left\langle \big(\frac{\partial h_\phi(x_i)}{\partial \phi}\big)^\top W^\top r_i, \big(\frac{\partial h_\phi(x_t)}{\partial \phi}\big)^\top W^\top r_t \right\rangle.$$

The first term is the Frobenius inner product between the two classifier gradients:

$$\big\langle r_i h_i^\top, r_t h_t^\top \big\rangle_F = \operatorname{tr}\left[(r_i h_i^\top)^\top (r_t h_t^\top)\right] = \operatorname{tr}\left[h_i r_i^\top r_t h_t^\top\right] = (r_i^\top r_t)(h_i^\top h_t).$$

For the representation-parameter term, we obtain

$$\begin{aligned}\left\langle \big(\frac{\partial h_\phi(x_i)}{\partial \phi}\big)^\top W^\top r_i, \big(\frac{\partial h_\phi(x_t)}{\partial \phi}\big)^\top W^\top r_t \right\rangle &= \left[\big(\frac{\partial h_\phi(x_i)}{\partial \phi}\big)^\top W^\top r_i\right]^\top \left[\big(\frac{\partial h_\phi(x_t)}{\partial \phi}\big)^\top W^\top r_t\right] \\ &= r_i^\top W \frac{\partial h_\phi(x_i)}{\partial \phi} \left(\frac{\partial h_\phi(x_t)}{\partial \phi}\right)^\top W^\top r_t.\end{aligned}$$

Combining the two contributions gives

$$\langle g_i, g_t \rangle = (r_i^\top r_t)(h_i^\top h_t) + r_i^\top W \frac{\partial h_\phi(x_i)}{\partial \phi} \big(\frac{\partial h_\phi(x_t)}{\partial \phi}\big)^\top W^\top r_t,$$

as in Eq. (5) in the main text.

C Implementation Details

This appendix outlines the experimental setup for the main evaluation in Section 4.1. Unless noted otherwise, RAND and BASIS use the same targets, poison budgets, attack hyperparameters, and victim training procedures. They differ only in the clean training examples chosen as poison bases.

C.1 DATASETS

All main experiments use CIFAR-10's standard train/test splits. Images are normalized per channel with $\mu = (0.4914, 0.4822, 0.4465)$ and $\sigma = (0.2023, 0.1994, 0.2010)$. Base selection and model training use normalized inputs, but selected bases are de-normalized to $[0, 1]$ for poison optimization. Perturbations are optimized within the allowed budget, and the resulting poisoned images are clipped to $[0, 1]$ after each step. The final poisoned images are re-normalized before insertion into the training set. Throughout the paper, class pairs follow the poison-class→target-class convention, $y_{\mathrm{adv}} \to y_t$. Thus, dog→bird means modifying dog training samples so that a bird is classified as a dog. Poisoned examples keep their original label y_{adv} during victim training.

C.2 MODEL TRAINING

We evaluate ConvNet, ResNet20, and VGG13. ConvNet consists of three convolutional stages with a width of 128, ReLU activations, batch normalization, and average pooling. ResNet20 and VGG13 also use their batch-normalized implementations. All surrogate and victim models are trained from scratch using cross-entropy loss and SGD with a momentum of 0.9. Each surrogate is trained for 60 epochs with a batch size of 128 and an initial learning rate of 0.1, reduced by a factor of 0.1 after epochs 35 and 45. No data augmentation is used during surrogate training. For each target, we train six independently initialized victim models from scratch on the poisoned training set. Victims are trained for 60 epochs with an initial learning rate of 0.1, reduced by a factor of 0.1 after epoch 40, a batch size of 125, and a momentum of 0.9. No victim-side data augmentation is used in the main evaluation.

C.3 TARGET SELECTION

Targets are selected from CIFAR-10 test examples belonging to the target class. We exclude examples that the ensemble of six victim models trained on the clean training set already predicts as y_{adv}, since these would constitute successful targeted predictions before poisoning. For each model, attack, and class pair, we evaluate ten targets. Once selected, the target indices are fixed and reused across all base-selection variants and poison budgets. RAND and BASIS therefore attack exactly the same target images. Target sets can differ between perturbation objectives. We therefore interpret aggregate ASR improvements as averages of within-configuration comparisons between BASIS and RAND, rather than as controlled comparisons between perturbation objectives. The target indices used in the main evaluation are reported in Table 6.

Table 6: Target images used in the main evaluation. The same ten CIFAR-10 test images are used across base-selection variants and poison budgets for each model, class pair, and attack.

Model	Pair	Attack	CIFAR-10 test indices
ConvNet	dog→bird	BP	2540, 8129, 3725, 5705, 7663, 3875, 5169, 6335, 7488, 9831
ConvNet	dog→bird	GM, SAPA	843, 1257, 8252, 8730, 8055, 5859, 8954, 821, 1547, 4427
ConvNet	frog→airplane	BP	7511, 857, 7010, 2091, 1776, 1596, 5354, 3158, 6006, 6191
ConvNet	frog→airplane	GM, SAPA	496, 2087, 8941, 9820, 3825, 1796, 2548, 6437, 3336, 2603
ResNet20	dog→bird	BP	3495, 2903, 6697, 2315, 6899, 2312, 6353, 4421, 2242, 6768
ResNet20	dog→bird	GM, SAPA	5118, 8740, 5324, 4630, 833, 2065, 7056, 9059, 6707, 9874
ResNet20	frog→airplane	BP	9273, 2535, 4118, 5462, 8476, 3342, 313, 4490, 2672, 6481
ResNet20	frog→airplane	GM, SAPA	9235, 338, 255, 3207, 8563, 912, 4639, 2570, 3117, 2041
VGG13	dog→bird	BP	3728, 5640, 6922, 7855, 2127, 7586, 7360, 2171, 4462, 2597
VGG13	dog→bird	GM, SAPA	630, 409, 2270, 9870, 1656, 5022, 9090, 6033, 7940, 9731
VGG13	frog→airplane	BP	912, 9957, 1739, 1165, 3278, 4313, 416, 6491, 338, 4287
VGG13	frog→airplane	GM, SAPA	2783, 2507, 6738, 5787, 7815, 5074, 7356, 8267, 2264, 6481

C.4 PERTURBATION OPTIMIZATION

We provide details for poisoning objectives used in our evaluation.

- **Bullseye Polytope (BP)** (Aghakhani et al., 2021): We minimize the distance between the target representation and the poison set's mean representation across crafting surrogates. BP runs for 250 sign-gradient steps with a step size of $1/255$. Perturbations are initialized uniformly within the feasible region and projected onto the $8/255$ constraint after each update.

- **Gradient Matching (GM)** (Geiping et al., 2021): We align the poison-set gradient with the target gradient under the adversarial label. For surrogate k let $g_t^{(k)}$ and $g_p^{(k)}$ be the target and poison gradients. We minimize the mean cosine mismatch:

$$\mathcal{L}_{\mathrm{GM}} = \frac{1}{K_c} \sum_{k=1}^{K_c} \left(1 - \frac{\langle g_p^{(k)}, g_t^{(k)} \rangle}{\|g_p^{(k)}\|_2 \, \|g_t^{(k)}\|_2} \right), \qquad K_c = 20.$$

  We optimize perturbations for 250 Adam steps with a learning rate $1/255$, supplying the elementwise sign of the perturbation gradient to Adam before its moment updates and projecting to the feasible set after each update.

- **Sharpness-Aware Poisoning (SAPA)** (He et al., 2024): SAPA uses the same poison optimization as GM (20 crafting surrogates, 250 steps) but replaces the standard target gradient with a sharpness-aware version. For each surrogate we compute the target gradient $g_t^{(k)}$ and evaluate the matching objective at perturbed model parameters.

$$\theta' = \theta + \gamma \frac{g_t^{(k)}}{\|g_t^{(k)}\|_2}, \qquad \gamma = 0.05.$$

D Additional Results and Statistical Analysis

The main evaluation in Table 1 considers ten attack targets, with six independently initialized victim models trained for each target. While the main table reports ASR, Figure 2 shows CTA only for a subset of the evaluated settings. We therefore report the complete CTA results in Table 7 and complement the main ASR results with a target-level statistical analysis in Table 9. We additionally recompute the paired statistics for nine selected configurations using complete target-level records and report them in Table 8. The complete CTA results show that the improvement in attack success does not come with a comparable reduction in clean performance. Across the 108 configurations, mean CTA is only 0.02 percentage points lower for BASIS than for RAND. Overall, clean accuracy remains largely unchanged despite substantially different attack success rates.

Table 7: CTA results corresponding to the ASR evaluation in Table 1. Entries report CTA (%, ↑).

		dog→bird						frog→airplane					
	ρ (%)	BP		GM		SAPA		BP		GM		SAPA	
		RAND	BASIS	RAND	BASIS	RAND	BASIS	RAND	BASIS	RAND	BASIS	RAND	BASIS
ConvNet	0.05	80.14	80.18	80.34	80.46	80.43	80.42	80.04	79.97	80.24	80.33	80.18	80.32
	0.1	80.14	80.18	80.34	80.29	80.43	80.32	80.04	80.09	80.24	80.24	80.18	80.19
	0.2	80.13	80.12	80.38	80.38	80.39	80.42	80.10	80.04	80.16	80.17	80.15	80.18
	0.5	80.11	80.22	80.46	80.41	80.30	80.40	80.06	80.04	80.20	80.03	80.16	80.18
	1	80.10	80.14	80.46	80.57	80.36	80.43	80.09	80.02	80.17	80.17	80.14	80.10
	2	80.09	80.11	80.38	80.40	80.31	80.39	80.02	79.91	80.10	80.05	80.10	80.05
ResNet20	0.05	83.75	83.73	83.82	83.77	83.73	83.73	83.77	83.82	83.82	83.80	83.77	83.77
	0.1	83.80	83.81	83.76	83.61	83.76	83.71	83.75	83.67	83.80	83.65	83.75	83.68
	0.2	83.73	83.68	83.88	83.79	83.63	83.78	83.76	83.74	83.77	83.79	83.79	83.77
	0.5	83.71	83.73	83.79	83.81	83.85	83.89	83.69	83.72	83.75	83.73	83.80	83.79
	1	83.62	83.62	83.73	83.70	83.74	83.76	83.86	83.72	83.72	83.69	83.68	83.78
	2	83.56	83.72	83.67	83.72	83.62	83.64	83.72	83.66	83.57	83.45	83.67	83.45
VGG13	0.05	85.44	85.42	85.52	85.59	85.57	85.53	85.45	85.40	85.21	85.18	85.43	85.23
	0.1	85.34	85.45	85.52	85.33	85.53	85.50	85.34	85.36	85.21	85.23	85.43	85.19
	0.2	85.25	85.30	85.54	85.46	85.59	85.55	85.34	85.47	85.23	85.37	85.26	85.25
	0.5	85.42	85.37	85.49	85.46	85.49	85.47	85.30	85.39	85.20	85.30	85.39	85.16
	1	85.30	85.30	85.45	85.52	85.36	85.29	85.36	85.26	85.35	85.19	85.21	85.12
	2	85.32	85.29	85.29	85.25	85.33	85.30	85.45	85.30	85.11	85.09	85.23	85.00

In the main evaluation, each target and selector is evaluated using one selected base set and one optimized poison set shared across six independently initialized victims, so the target is the unit of analysis. The confidence intervals are conditional on the realized surrogate ensembles and poison sets; they do not separately quantify variability from independently repeating base selection, surrogate training, or poison optimization for the same targets. For method v, we compute the success

rate $r_t^{(v)}$ as the average of six binary outcomes for target t, where $a_{t,s}^{(v)} = 1$ if the attack succeeds and 0 otherwise. ASR is the mean of $r_t^{(v)}$ over the ten targets. In Table 8, the standard error is $\mathrm{SE}^{(v)} = \mathrm{SD}(r_1^{(v)}, \ldots, r_T^{(v)})/\sqrt{T}$, where $T = 10$ and SD uses the sample denominator $T - 1$. Because RAND and BASIS are evaluated on the same targets, we compute their paired success-rate difference d_t for each target and average these differences to obtain Δ. ASR and SE are reported as percentages (multiplied by 100), and 100Δ is the paired difference in percentage points. These are defined as

$$r_t^{(v)} = \frac{1}{6}\sum_{s=1}^{6} a_{t,s}^{(v)}, \qquad \mathrm{ASR}^{(v)} = \frac{1}{10}\sum_{t=1}^{10} r_t^{(v)}, \qquad d_t = r_t^{(\mathrm{Basis})} - r_t^{(\mathrm{Rand})}, \qquad \Delta = \frac{1}{10}\sum_{t=1}^{10} d_t.$$

Thus, 100Δ gives the average ASR improvement of BASIS over RAND in percentage points. The 95% confidence interval in Table 9 quantifies uncertainty in this paired difference and reflects how it might vary across different samples of attack targets. An interval entirely above zero supports an improvement for BASIS, while an interval entirely below zero favors RAND. Intervals containing zero are consistent with no difference. We estimate these intervals using a paired bootstrap over targets. Each replicate samples ten targets with replacement, retaining all six victim outcomes for each selected target under both methods. This preserves the grouping of victim runs and the pairing between RAND and BASIS. We recompute 100Δ for each replicate and use the 2.5th and 97.5th percentiles of the resulting bootstrap distribution as the 95% confidence interval in percentage points.

Table 9 shows that the average attack success increases from 20.3% for RAND to 44.7% for BASIS. Among 108 configurations, 69 confidence intervals lie entirely above zero, 39 include zero, and none are completely negative. For GM and SAPA, 27 of 36 and 29 of 36 intervals are entirely positive. One ResNet20 configuration with BP slightly favors RAND, but its confidence interval includes zero. Table 8 further reports recomputed statistics for nine representative configurations spanning all architectures, attack methods, both class pairs, and multiple poisoning budgets. These pointwise intervals serve as a targeted robustness check rather than a multiple-comparison-corrected analysis over all 108 configurations. We use 20000 paired bootstrap resamples across ten targets and six victim-training runs to generate unadjusted pointwise 95% confidence intervals.

The seven evaluated GM and SAPA configurations exhibit observed gains of 8.3 to 41.7 percentage points, with every pointwise interval residing entirely above zero. Conversely, the two BP configurations show observed gains of $+8.3$ and $+15.0$ percentage points, respectively, with both confidence intervals including zero. These results support improvements in the seven audited GM and SAPA configurations; the two audited BP confidence intervals include zero. The recomputed statistics for these nine configurations are also incorporated into Table 9.

Table 8: Paired target-level ASR analysis. Both selectors use the same ten targets and six victim runs per target. Entries report ASR (%) with target-level SE in parentheses. 100Δ is the BASIS–RAND difference before rounding. SE, 100Δ, and pointwise 95% confidence limits are expressed in percentage points.

Model	Class Pair	Attack	ρ (%)	Targets	ASR (SE)		100Δ	Paired 95% CI
					RAND	BASIS		
ConvNet	dog → bird	SAPA	0.1	10	0.0 (0.0)	21.7 (5.6)	+21.7	[11.7, 31.7]
	dog → bird	SAPA	0.2	10	0.0 (0.0)	26.7 (7.5)	+26.7	[15.0, 41.7]
	frog → airplane	SAPA	0.2	10	0.0 (0.0)	8.3 (3.7)	+8.3	[1.7, 15.0]
ResNet20	dog → bird	GM	0.5	10	43.3 (7.5)	76.7 (7.9)	+33.3	[21.7, 45.0]
	frog → airplane	GM	0.2	10	0.0 (0.0)	36.7 (6.9)	+36.7	[23.3, 50.0]
	frog → airplane	BP	0.2	10	8.3 (3.7)	16.7 (3.5)	+8.3	[−1.7, 16.7]
VGG13	dog → bird	GM	0.5	10	55.0 (9.3)	95.0 (3.6)	+40.0	[21.7, 58.3]
	frog → airplane	GM	0.2	10	5.0 (3.6)	46.7 (10.8)	+41.7	[20.0, 63.3]
	dog → bird	BP	0.2	10	21.7 (10.3)	36.7 (6.9)	+15.0	[−8.3, 36.7]

Table 9: Target-level statistical analysis of the ASR results in Table 1. Entries report mean ASR (%) across ten targets with six victim runs per target, with target-level SEs in parentheses. The 95% CIs are paired bootstrap intervals for the BASIS–RAND difference, in percentage points.

		BP			GM			SAPA		
	ρ (%)	RAND ASR (SE)	BASIS ASR (SE)	95% CI BASIS–RAND	RAND ASR (SE)	BASIS ASR (SE)	95% CI BASIS–RAND	RAND ASR (SE)	BASIS ASR (SE)	95% CI BASIS–RAND
ConvNet										
dog→bird	0.05	0.0 (0.0)	13.3 (5.4)	[2.5,22.5]	0.0 (0.0)	8.3 (5.1)	[0.0,17.5]	0.0 (0.0)	13.3 (6.5)	[2.5,25.0]
	0.1	0.0 (0.0)	28.3 (5.6)	[18.3,38.3]	0.0 (0.0)	11.7 (5.6)	[0.0,22.2]	0.0 (0.0)	21.7 (5.6)	[11.7,31.7]
	0.2	1.7 (1.7)	65.0 (7.2)	[50.0,75.0]	0.0 (0.0)	28.3 (6.6)	[16.7,41.7]	0.0 (0.0)	26.7 (7.5)	[15.0,41.7]
	0.5	13.3 (6.5)	88.3 (4.3)	[58.3,88.3]	1.7 (1.7)	73.3 (9.4)	[52.1,87.5]	6.7 (3.7)	48.3 (11.8)	[20.0,63.3]
	1	35.0 (8.0)	96.7 (2.2)	[46.7,75.0]	21.7 (7.9)	86.7 (7.4)	[40.8,79.6]	23.3 (7.5)	81.7 (10.1)	[41.7,71.7]
	2	73.3 (9.7)	98.3 (1.7)	[10.0,45.0]	43.3 (9.4)	95.0 (5.0)	[26.3,52.5]	50.0 (10.5)	100.0 (0.0)	[22.9,62.5]
frog→airplane	0.05	0.0 (0.0)	5.0 (5.0)	[0.0,15.0]	0.0 (0.0)	0.0 (0.0)	[0.0,0.0]	0.0 (0.0)	3.3 (2.2)	[0.0,7.5]
	0.1	0.0 (0.0)	5.0 (3.6)	[0.0,13.3]	0.0 (0.0)	3.3 (2.2)	[0.0,7.5]	0.0 (0.0)	3.3 (3.3)	[0.0,10.0]
	0.2	0.0 (0.0)	8.3 (3.7)	[1.7,15.0]	0.0 (0.0)	3.3 (2.2)	[0.0,8.3]	0.0 (0.0)	8.3 (3.7)	[1.7,15.0]
	0.5	0.0 (0.0)	26.7 (6.7)	[15.0,40.0]	0.0 (0.0)	5.0 (2.5)	[0.0,10.4]	0.0 (0.0)	5.0 (2.5)	[0.0,10.0]
	1	1.7 (1.7)	36.7 (10.2)	[16.7,53.3]	3.3 (2.2)	20.0 (9.9)	[2.5,33.3]	1.7 (1.7)	21.7 (8.3)	[6.7,35.0]
	2	10.0 (4.4)	55.0 (13.8)	[21.7,68.3]	3.3 (3.3)	35.0 (7.6)	[18.3,43.3]	8.3 (5.1)	25.0 (9.0)	[6.7,22.1]
ResNet20										
dog→bird	0.05	1.7 (1.7)	15.0 (7.2)	[−2.5,30.0]	6.7 (5.1)	15.0 (6.3)	[−10.0,25.0]	6.7 (3.7)	28.3 (12.4)	[−2.5,47.5]
	0.1	11.7 (5.0)	18.3 (8.0)	[−7.1,12.5]	15.0 (5.2)	35.0 (11.2)	[3.8,37.9]	18.3 (6.3)	43.3 (5.7)	[20.4,39.2]
	0.2	16.7 (5.0)	25.0 (6.7)	[0.0,16.7]	31.7 (8.4)	56.7 (13.2)	[1.7,62.5]	21.7 (7.9)	53.3 (12.6)	[18.8,57.9]
	0.5	30.0 (7.8)	46.7 (11.1)	[1.7,31.7]	43.3 (7.5)	76.7 (7.9)	[21.7,45.0]	55.0 (10.6)	78.3 (12.2)	[4.6,55.4]
	1	43.3 (10.3)	55.0 (12.7)	[−8.3,35.0]	56.7 (11.4)	91.7 (5.1)	[20.8,60.8]	65.0 (8.8)	85.0 (6.3)	[4.6,48.3]
	2	55.0 (8.3)	48.3 (13.7)	[−33.3,2.1]	75.0 (8.7)	95.0 (5.0)	[10.4,41.7]	95.0 (3.6)	100.0 (0.0)	[0.0,11.7]
frog→airplane	0.05	5.0 (5.0)	3.3 (2.2)	[−15.0,7.5]	1.7 (1.7)	5.0 (3.6)	[0.0,7.5]	0.0 (0.0)	15.0 (8.0)	[2.5,32.5]
	0.1	8.3 (3.7)	5.0 (2.5)	[−16.7,3.8]	1.7 (1.7)	23.3 (8.7)	[7.5,35.8]	1.7 (1.7)	28.3 (9.3)	[10.4,42.9]
	0.2	8.3 (3.7)	16.7 (3.5)	[−1.7,16.7]	0.0 (0.0)	36.7 (6.9)	[23.3,50.0]	5.0 (3.6)	20.0 (7.8)	[−2.5,35.0]
	0.5	18.3 (5.8)	25.0 (9.4)	[−7.4,25.9]	8.3 (3.7)	60.0 (7.9)	[33.3,68.3]	3.3 (2.2)	50.0 (11.4)	[22.9,66.7]
	1	16.7 (5.0)	40.0 (10.0)	[2.1,49.2]	26.7 (9.7)	75.0 (9.0)	[25.8,66.7]	21.7 (6.6)	66.7 (9.3)	[36.7,62.1]
	2	30.0 (10.2)	41.7 (12.5)	[−18.8,37.5]	45.0 (8.6)	90.0 (5.7)	[22.9,56.3]	40.0 (7.1)	83.3 (6.1)	[37.5,60.4]
VGG13										
dog→bird	0.05	21.7 (7.0)	35.0 (10.4)	[−10.0,32.5]	0.0 (0.0)	38.3 (13.4)	[15.0,62.5]	0.0 (0.0)	28.3 (6.6)	[15.0,40.0]
	0.1	21.7 (6.1)	28.3 (10.6)	[−2.9,14.2]	13.3 (6.0)	53.3 (9.2)	[26.7,45.8]	11.7 (5.6)	53.3 (12.6)	[15.8,65.8]
	0.2	21.7 (10.3)	36.7 (6.9)	[−8.3,36.7]	21.7 (9.3)	56.7 (11.7)	[14.6,47.9]	26.7 (9.0)	83.3 (7.9)	[34.6,69.6]
	0.5	45.0 (7.9)	51.7 (10.7)	[−8.3,23.3]	55.0 (9.3)	95.0 (3.6)	[21.7,58.3]	60.0 (9.7)	95.0 (3.6)	[8.8,53.3]
	1	41.7 (8.7)	53.3 (11.3)	[−11.7,30.0]	76.7 (10.0)	96.7 (3.3)	[2.1,35.4]	88.3 (5.6)	100.0 (0.0)	[2.1,16.7]
	2	61.7 (11.7)	65.0 (10.7)	[−8.3,12.5]	96.7 (3.3)	100.0 (0.0)	[0.0,10.0]	100.0 (0.0)	100.0 (0.0)	[0.0,0.0]
frog→airplane	0.05	1.7 (1.7)	3.3 (2.2)	[−7.5,7.5]	0.0 (0.0)	10.0 (5.1)	[2.5,20.0]	0.0 (0.0)	18.3 (10.4)	[2.5,40.0]
	0.1	1.7 (1.7)	8.3 (5.1)	[0.0,12.9]	0.0 (0.0)	40.0 (8.7)	[25.0,55.0]	0.0 (0.0)	38.3 (12.7)	[15.0,62.5]
	0.2	1.7 (1.7)	3.3 (2.2)	[−3.3,6.7]	5.0 (3.6)	46.7 (10.8)	[20.0,63.3]	10.0 (4.4)	48.3 (12.5)	[19.2,55.4]
	0.5	3.3 (2.2)	8.3 (4.5)	[−3.3,13.3]	16.7 (6.6)	66.7 (12.7)	[30.0,70.0]	15.0 (6.8)	66.7 (8.6)	[31.2,66.7]
	1	13.3 (4.8)	30.0 (12.6)	[1.7,48.3]	28.3 (8.6)	83.3 (11.4)	[33.3,77.1]	20.0 (8.5)	85.0 (10.7)	[40.8,85.8]
	2	10.0 (6.7)	16.7 (9.0)	[−2.1,31.2]	56.7 (10.3)	88.3 (10.0)	[14.6,45.8]	53.3 (9.9)	86.7 (10.2)	[14.6,47.9]

E GENERALIZATION ACROSS CLASS PAIRS AND DATASETS

The main evaluation in Section 4.1 considers two class pairs. Here, we first examine whether the benefit of target-conditioned base selection extends to a broader set of CIFAR-10 attack directions and then evaluate its generalization to additional datasets. Table 10 evaluates six additional class pairs under the same poisoning budget and victim architecture, together with dog→bird as a reference. Across the six additional pairs, BASIS increases mean ASR from 12.7% to 64.7% for BP, from 12.0% to 46.7% for GM, and from 14.0% to 44.7% for SAPA. Also, several settings with weak random initialization become highly successful after base selection, whereas other pairs remain difficult even with BASIS. This variation indicates that base selection does not remove the inherent difficulty of particular target directions, but its benefit is not specific to the two class pairs used in the main evaluation.

Table 10: Generalization to additional CIFAR-10 class pairs using ConvNet at $\rho = 0.005$. Each setting uses five targets and five trained victims per target. Entries report ASR (%, ↑).

	BP		GM		SAPA	
Poison → Target	**RAND**	**BASIS**	**RAND**	**BASIS**	**RAND**	**BASIS**
dog → bird	24.0	80.0	12.0	64.0	24.0	68.0
dog → cat	40.0	100.0	44.0	80.0	36.0	80.0
horse → deer	8.0	100.0	4.0	72.0	4.0	72.0
truck → automobile	4.0	68.0	12.0	48.0	12.0	48.0
airplane → bird	24.0	88.0	12.0	80.0	32.0	64.0
frog → ship	0.0	4.0	0.0	0.0	0.0	0.0
cat → truck	0.0	28.0	0.0	0.0	0.0	4.0

We next evaluate whether the benefit of target-conditioned base selection in improving attack success extends beyond CIFAR-10. Table 11 considers CIFAR-100, SVHN, and Tiny ImageNet using dataset-appropriate victim architectures and multiple poisoning objectives. The evaluation includes both a budget sweep for one class pair and additional class pairs at a fixed budget. The advantage of BASIS remains visible under these changes. On CIFAR-100, it improves or matches RAND in 20 of the 21 evaluated attack settings, although the magnitude of the improvement varies across objectives and class pairs. The effect is particularly consistent on SVHN, where BASIS achieves higher ASR in all 24 comparisons. The single Tiny ImageNet pair provides a more limited test, but BASIS improves BP and GM while matching RAND under SAPA. CTA remains broadly comparable between the two selection methods, indicating that the increase in attack success is not associated with a systematic loss in clean predictive performance. Taken together, these results show that the advantage of informed base selection persists across changes in class pair and dataset.

Table 11: Generalization across CIFAR-100, SVHN, and Tiny ImageNet. We compare RAND and BASIS. Entries report ASR (%) with CTA (%) in parentheses.

Dataset **Model**	**Poison → Target**	ρ (%)	BP		GM		SAPA	
			RAND	BASIS	RAND	BASIS	RAND	BASIS
CIFAR-100 ResNet18	sea → willow tree	0.1	6.3 (57.90)	6.3 (57.83)	12.5 (57.16)	37.5 (57.82)	12.5 (57.94)	25.0 (57.77)
	sea → willow tree	0.2	0.0 (57.84)	6.3 (57.58)	43.8 (57.98)	43.8 (57.03)	50.0 (57.66)	50.0 (57.53)
	sea → willow tree	0.5	0.0 (57.89)	12.5 (57.98)	93.8 (57.47)	100.0 (58.26)	81.3 (57.37)	75.0 (57.36)
	plain → bicycle	0.2	0.0 (58.04)	0.0 (56.98)	0.0 (58.09)	12.5 (57.74)	0.0 (57.56)	0.0 (57.88)
	wardrobe → lawn mower	0.2	6.3 (58.16)	6.3 (57.46)	6.3 (58.13)	25.0 (58.05)	0.0 (58.03)	18.8 (57.29)
	bottle → road	0.2	0.0 (57.80)	0.0 (57.97)	0.0 (57.57)	31.2 (57.39)	0.0 (57.16)	12.5 (57.58)
	sunflower → cattle	0.2	0.0 (57.52)	6.3 (58.05)	6.3 (57.16)	25.0 (57.91)	0.0 (57.74)	12.5 (58.38)
SVHN ConvNet	9 → 2	0.1	3.3 (92.18)	33.3 (92.19)	0.0 (92.20)	63.3 (92.18)	3.3 (92.23)	50.0 (92.16)
	9 → 2	0.2	16.7 (92.15)	43.3 (92.20)	16.7 (92.17)	56.7 (92.18)	13.3 (92.16)	73.3 (92.18)
	9 → 2	0.5	20.0 (92.17)	36.7 (92.17)	40.0 (92.21)	80.0 (92.16)	20.0 (92.19)	83.3 (92.15)
	9 → 2	1	20.0 (92.15)	33.3 (92.14)	50.0 (92.18)	83.3 (92.11)	53.3 (92.18)	83.3 (92.14)
	6 → 1	0.2	6.7 (92.18)	56.7 (92.19)	13.3 (92.18)	90.0 (92.17)	13.3 (92.19)	90.0 (92.20)
	8 → 3	0.2	10.0 (92.21)	66.7 (92.19)	26.7 (92.19)	60.0 (92.18)	30.0 (92.19)	63.3 (92.17)
	7 → 5	0.2	0.0 (92.22)	23.3 (92.21)	0.0 (92.18)	50.0 (92.19)	0.0 (92.22)	53.3 (92.16)
	0 → 4	0.2	13.3 (92.21)	20.0 (92.20)	16.7 (92.20)	50.0 (92.20)	16.7 (92.21)	50.0 (92.19)
Tiny ImageNet ResNet18	`n01443537` → `n01629819`	0.2	0.0 (49.18)	12.5 (49.18)	0.0 (49.41)	6.3 (49.41)	0.0 (49.50)	0.0 (49.50)

F ADDITIONAL EVALUATION FOR ARCHITECTURE MISMATCH

F.1 FULL ARCHITECTURE MISMATCH BETWEEN ATTACK AND VICTIM SURROGATES

Section 4.2 evaluates the transferability of BASIS by varying the selection architecture while keeping the perturbation optimization surrogates matched to the victim. In this section, we extend that analysis to a full architecture mismatch scenario. Here, the attacker relies on a single surrogate architecture A for both base sample selection and the subsequent poison perturbation optimization while the victim model is trained on a different architecture V.

Table 12 reports the results of this fully mismatched setting for the GM and SAPA objectives. As expected, the overall attack success rate decreases when the perturbation optimization surrogates

Table 12: Full architecture mismatch results on CIFAR-10 dog→bird. Rows identify the architecture A used for BASIS selection and poison construction; column groups identify the victim architecture V. The $K = 10$ and $K = 20$ columns report BASIS using K selection surrogates and 20 crafting surrogates. Entries report ASR (%). Shared RAND values are reference baselines shown once per block, not architecture-specific controls. Shaded cells indicate matched architectures, $A = V$.

Attack	ρ (%)	Surrogate architecture	ConvNet			ResNet20			VGG13		
			RAND	$K = 10$	$K = 20$	RAND	$K = 10$	$K = 20$	RAND	$K = 10$	$K = 20$
GM	0.2	ConvNet	0.0	28.3	28.3	31.7	6.7	10.0	21.7	33.3	28.3
		ResNet20		11.7	16.7		46.7	58.3		48.3	48.3
		VGG13		6.7	5.0		35.0	36.7		70.0	80.0
	0.5	ConvNet	1.7	46.7	50.0	43.3	15.0	13.3	55.0	40.0	46.7
		ResNet20		18.3	33.3		76.7	76.7		63.3	60.4
		VGG13		31.7	36.7		72.9	70.8		95.0	95.0
SAPA	0.2	ConvNet	0.0	28.3	26.7	21.7	11.7	10.0	26.7	33.3	23.3
		ResNet20		18.3	11.7		55.0	61.7		51.7	38.3
		VGG13		13.3	6.7		38.3	51.7		78.3	83.3
	0.5	ConvNet	6.7	46.7	48.3	55.0	21.7	15.0	60.0	45.0	55.0
		ResNet20		28.3	41.7		78.3	78.3		63.8	72.3
		VGG13		30.0	28.3		70.8	68.8		96.7	97.9

diverge from the victim architecture. Despite the compounded difficulty of this setting, BASIS continues to produce successful attacks and frequently exceeds the displayed Rand reference. Attacks crafted entirely using deeper surrogate architectures, such as ResNet20 and VGG13, exhibit stronger transferability across different victims. Conversely, attacks generated solely with the shallower ConvNet architecture transfer less effectively to deeper victim models. These results indicate that while a full architecture mismatch degrades absolute attack strength, BASIS still produces successful attacks in the evaluated cross-architecture settings.

F.2 EFFECT OF SELECTION ARCHITECTURE UNDER PAIRED EVALUATION

Section 4.2 and Table 2 compare selection architectures and ensemble sizes using different runs for the matched and mismatched averages. Here, we pair the evaluations to assess the effect of selection architecture. We evaluate GM and SAPA on CIFAR-10 for dog→bird using ConvNet for poison crafting and victim training. BASIS selects bases with $K = 20$ ConvNet, ResNet20, or VGG13 surrogates, so architecture mismatch affects only selection. All selection architectures use the same targets, crafting settings, and victim-training seeds, with a shared RAND baseline. ASR is averaged over six targets and six victim runs per target. Table 13 shows that BASIS achieves higher ASR than the shared RAND baseline in 17 of 18 comparisons and matches it in the remaining comparison. At $\rho = 0.005$, ResNet20 selection achieves the highest observed ASR for both GM and SAPA. However, no selection architecture performs best throughout.

Matched ConvNet selection achieves higher ASR than either mismatched selector for SAPA at the two smaller budgets. To assess the effect of changing the selection architecture, we compute paired ASR differences relative to matched ConvNet selection. For each attack and budget, let $r_{t,S}$ denote the success rate across victim-training runs for target t and selection architecture S. We use targets as the statistical units and report the mean paired difference $\bar{d}_S$ with a pointwise 95% paired t-confidence interval. Let $s_{d,S}$ denote the sample standard deviation of target-level differences. The differences and CIs are expressed in percentage points and computed as

$$d_{t,S} = 100\big(r_{t,S} - r_{t,\mathrm{ConvNet}}\big), \qquad \bar{d}_S = \frac{1}{6}\sum_{t=1}^{6} d_{t,S}, \qquad \mathrm{CI}_{95\%,S} = \bar{d}_S \pm t_{0.975,5}\frac{s_{d,S}}{\sqrt{6}}.$$

With these factors held fixed, the paired comparisons provide a more direct assessment of selection-architecture effects than the broader results in Table 2. Under this shared protocol, gains over RAND persist when selection architectures differ from the victim, while poison crafting remains victim matched. However, all 12 confidence intervals in Table 13 include zero and allow substantial increases or decreases in ASR. The experiment therefore supports the usefulness of cross-architecture base selection in the evaluated settings, but the six-target comparison does not establish equivalent performance or a negligible effect of architecture mismatch.

Table 13: Selection architectures on CIFAR-10, dog→bird, with ConvNet crafting and victims. Entries report ASR (%), with gains over RAND in percentage points in parentheses. Gains and paired differences are computed before rounding. Bold marks row maxima. Rightmost columns give differences from ConvNet selection and pointwise 95% CIs, in percentage points.

			BASIS selection architecture S			ResNet20 − ConvNet		VGG13 − ConvNet	
Attack	ρ (%)	RAND	ConvNet	ResNet20	VGG13	$\bar{d}_S$	95% CI	$\bar{d}_S$	95% CI
GM	0.1	2.8	**8.3** (+5.6)	2.8 (+0.0)	5.6 (+2.8)	−5.6	[−23.6, 12.5]	−2.8	[−20.0, 14.4]
	0.2	0.0	19.4 (+19.4)	**22.2** (+22.2)	**22.2** (+22.2)	+2.8	[−17.7, 23.2]	+2.8	[−14.4, 20.0]
	0.5	11.1	47.2 (+36.1)	**50.0** (+38.9)	38.9 (+27.8)	+2.8	[−14.4, 20.0]	−8.3	[−37.1, 20.4]
SAPA	0.1	0.0	**19.4** (+19.4)	5.6 (+5.6)	5.6 (+5.6)	−13.9	[−34.3, 6.6]	−13.9	[−34.3, 6.6]
	0.2	0.0	**27.8** (+27.8)	19.4 (+19.4)	8.3 (+8.3)	−8.3	[−32.4, 15.8]	−19.4	[−39.9, 1.0]
	0.5	2.8	36.1 (+33.3)	**38.9** (+36.1)	33.3 (+30.6)	+2.8	[−32.9, 38.5]	−2.8	[−40.2, 34.6]

G ADDITIONAL EVALUATION UNDER AUGMENTATION AND DEFENSES

G.1 DEFENSE EVALUATION WITH CALIBRATED CLEAN ACCURACY

We extend Section 4.3 by calibrating defense strength to match clean accuracy. Using clean CIFAR-10 data and no attack feedback, we tune EPIC's retained training fraction and FRIENDS' noise magnitude to minimize degradation compared to an undefended model. This setup compares a realistic defense configuration tuned to preserve clean predictive utility. Table 14 indicates that BASIS retains an advantage for both attacks on dog→bird under both calibrated defenses. Results on frog→airplane vary by defense strategy. EPIC completely suppresses both selectors, whereas BASIS retains marginal attack success against FRIENDS. The benefit of BASIS generally persists when defense strength is constrained by clean utility, although an adequately tuned defense can still neutralize both selection strategies.

Table 14: Defense evaluation with calibrated CTA on CIFAR-10 (ConvNet, $\rho = 2\%$). Settings are tuned on clean data and fixed across poisoned runs. Entries report ASR (%).

			dog → bird		**frog → airplane**	
Attack	**Defense**	**CTA**	RAND	BASIS	RAND	BASIS
GM	EPIC	80.33	36.0	**52.0**	0.0	0.0
	FRIENDS	80.02	28.0	**72.0**	0.0	**8.0**
SAPA	EPIC	80.33	36.0	**64.0**	0.0	0.0
	FRIENDS	80.02	32.0	**80.0**	0.0	**20.0**

G.2 AUGMENTATION-AWARE POISON OPTIMIZATION

We examine whether the advantage of BASIS persists when poison optimization directly accounts for the augmentation applied during victim training. In the unmatched setting, augmentation occurs only during victim training. In the matched setting, poison optimization also incorporates the same augmentation distribution. Both RAND and BASIS use identical targets and poisoning budgets to ensure a fair comparison. The "–" entries for matched RandAugment indicate that these results are unavailable because the RandAugment transformations used here are not differentiable through the poison optimization procedure.

Table 15 shows that BASIS maintains its advantage under Crop+Flip with matched and unmatched optimization. Matching the augmentation distribution does not consistently raise ASR, suggesting that the selection advantage is not solely due to augmentation mismatch. With RandAugment, ASR remains low. BASIS improves GM but not SAPA, so selection benefits depend on the attack and augmentation.

Table 15: Augmentation-aware poison optimization on CIFAR-10 using ConvNet, dog→bird, and $\rho = 0.5\%$. Unmatched applies augmentation only during victim training while Matched applies the same augmentation during poison optimization. Entries report ASR (%).

		Unmatched		Matched	
Attack	**Augmentation**	RAND	BASIS	RAND	BASIS
GM	Crop+Flip	4.0	20.0	4.0	16.0
	RandAugment	0.0	8.0	–	
SAPA	Crop+Flip	0.0	16.0	0.0	16.0
	RandAugment	4.0	4.0	–	

H COMPARISON OF ALTERNATIVE SELECTION SCORES

We compare scoring alternatives with RAND and BASIS under matched experimental settings. Recall that the margin used by BASIS, $M_i = \max_{c \neq y_{\text{adv}}} z_c(x_i) - z_{y_{\text{adv}}}(x_i)$, is target-independent and considers the strongest competing class. As an alternative, we evaluate a target-specific margin $T_i = z_{y_t}(x_i) - z_{y_{\text{adv}}}(x_i)$. The remaining alternatives include classifier-only alignment $(r_i^\top r_t)(h_i^\top h_t)$, its normalized variant $(r_i^\top r_t)R_i$, the combination $\widetilde{R}_i + \widetilde{T}_i$, and confidence-only selection. Table 16 shows that no selector performs best throughout. BASIS achieves the highest ASR in five of the seven evaluated configurations, while alternative criteria perform best in the remaining settings. Among the new alternatives, normalized classifier alignment performs well for GM, while the target-specific margin performs best for SAPA. Adding similarity to the target-specific margin does not consistently improve ASR, indicating that the benefit of each scoring component depends on the attack and poisoning budget. A practical advantage of BASIS is that it remains computationally efficient and can be used as a plug-and-play selection strategy without modifying the underlying attack.

Table 16: Scoring alternatives on CIFAR-10 using ConvNet and dog→bird. Entries report ASR (%); budget ρ is in percentage. The new alternatives use eight targets and six victim runs per target. The highest and second-highest reported ASRs in each column are **bold** and underlined, including ties.

		BP		GM			SAPA	
Method	**Selection Rule**	0.1	0.2	0.1	0.2	0.5	0.1	0.2
RAND		0.0	1.7	0.0	0.0	1.7	0.0	0.0
Classifier-only alignment	$(r_i^\top r_t)(h_i^\top h_t)$	29.2	62.5	<u>25.0</u>	29.2	66.7	16.7	31.3
Normalized classifier alignment	$(r_i^\top r_t)R_i$	27.1	64.6	**29.2**	29.2	<u>75.0</u>	22.9	<u>37.5</u>
Target-class margin	$T_i = z_{y_t}(x_i) - z_{y_{\text{adv}}}(x_i)$	25.0	45.8	20.8	<u>33.3</u>	60.4	<u>25.0</u>	**39.6**
Similarity + target-class margin	$\widetilde{R}_i + \widetilde{T}_i$	25.0	<u>68.8</u>	22.9	<u>33.3</u>	54.2	16.7	<u>37.5</u>
Confidence-only	$-p_\theta(y_{\text{adv}} \mid x_i)$	<u>31.3</u>	47.9	22.9	14.6	54.2	12.5	31.3
BASIS		**33.3**	**70.8**	20.8	**41.7**	**79.2**	**29.2**	31.3

I SENSITIVITY TO THE MARGIN WEIGHT

We examine how the relative weighting of representation similarity and logit margin affects attack success using $s_i = \widetilde{R}_i + \lambda \widetilde{M}_i$, where $\lambda = 1$ corresponds to the default BASIS score. We evaluate six weights on CIFAR-10 using ConvNet, the dog→bird class pair, and GM and SAPA. Table 17 reports the resulting ASR, including a rerun of the default setting. We use $\lambda = 1$ as an untuned default, assigning equal weight to the standardized similarity and margin scores. Although victim-model evaluations are unavailable to the attacker, tuning on held-out surrogates would be possible. Other weights yield higher ASR in some evaluated configurations. Every tested weight outperforms RAND in all six evaluated configurations, and the default achieves a mean ASR of 34.7%, compared with 1.4% for random selection. These results support equal weighting as a practical default and show that substantial improvements over random selection can be obtained without tuning the weight using victim-model outcomes.

Table 17: Margin-weight sensitivity on CIFAR-10 using ConvNet on dog→bird. Entries report ASR (%), with row maxima in bold.

Attack	ρ (%)	RAND	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 5$	$\lambda = 10$	$\lambda = 200$	$\lambda = 1000$
GM	0.1	0.0	11.1	11.1	**26.4**	23.6	13.9	18.1
	0.2	0.0	15.3	27.8	**30.6**	**30.6**	29.2	**30.6**
	0.5	1.7	34.7	**72.2**	61.1	59.7	61.1	52.8
SAPA	0.1	0.0	12.5	22.2	23.6	**27.8**	16.7	22.2
	0.2	0.0	13.9	27.8	**38.9**	25.0	23.6	27.8
	0.5	6.7	25.0	47.2	59.7	**68.1**	63.9	48.6

J EFFECT OF BASE REDUNDANCY

We test whether concentrating a fixed poisoning budget on a few high-scoring candidates is better than scoring and selecting each sample independently. For Top-r, we select the top-r samples via Algorithm 1, then fill the remaining budget with their nearest eligible neighbors in representation space without replacement. Table 18 shows that concentrating the budget around high-scoring seeds improves upon RAND, but none of the Top-r variants matches BASIS. Performance also does not consistently improve as r increases. This indicates that proximity to a high-scoring sample is not sufficient. Evaluating and selecting every base according to its own score is more effective than filling the budget with neighbors of a smaller seed set.

Table 18: Impact of sample redundancy on CIFAR-10 with ConvNet, SAPA, dog→bird, and $\rho = 0.5\%$. Each setup includes m unique samples. Cells show mean ASR (%) over 5 independent trials.

	RAND	Top-1	Top-2	Top-5	Top-10	BASIS
ASR (%)	24.0	52.0	52.0	32.0	48.0	68.0

K COMPUTATIONAL COST OF BASE SELECTION

We measure the cost of feature comparison R, the logit margin M, and subset selection on an H100 GPU. We use CIFAR-10 with $K = 20$ pretrained surrogates and $\rho = 10^{-3}$, corresponding to $m = 50$ selected bases. For each architecture, we evaluate three targets from each of the dog→bird and frog→airplane class pairs, with five timing repeats per target after an untimed warm-up. This yields 30 measured calls per architecture. These candidate-scoring measurements exclude surrogate training and loading, poison optimization, and victim training. The subset-selection stage sorts candidates by their scores and returns the top m bases.

We additionally measure the full training cost of one surrogate on CIFAR-10 over five training runs per architecture. Table 19 reports these costs alongside candidate scoring. Training time and peak GPU memory are averaged across the five runs. With the surrogate ensemble already loaded, the recorded scoring stages take approximately 0.72, 1.00, and 1.54 seconds per target for ConvNet, ResNet20, and VGG13, respectively. Subset selection contributes 8.6–38.3 milliseconds. These subtotals summarize the measured stages and are not separately measured end-to-end runtimes.

When all $K = 20$ selection surrogates must be trained, multiplying the per-model training times by twenty gives estimated sequential training costs of 14.4, 34.7, and 31.1 minutes for ConvNet, ResNet20, and VGG13, respectively. These estimates exclude checkpoint loading. Crafting checkpoints can be reused for selection when their architecture and training setup are compatible, reducing the number of additional models that must be trained. A selection ensemble can also be reused across targets from the same dataset, spreading its training cost across attacks. The additional cost of BASIS therefore includes candidate scoring together with any surrogate training and loading required beyond the models already available for perturbation optimization. The reduction in poisoning budget measures savings in the number of modified samples. Its effect on total attacker runtime also depends on surrogate reuse and poison optimization, and is not quantified by these scoring measurements.

Table 19: Scoring and surrogate-training costs. Scoring time reports mean ± SD over 30 calls, in seconds per target, with a subtotal across stages. Scoring peak memory is in MB. Single-surrogate training time (s) and peak memory (GiB) are averaged over five runs.

Measurement	ConvNet	ResNet20	VGG13
Candidate scoring with $K = 20$ surrogates			
Feature comparison R (s)	0.3438 ± 0.0003	0.5017 ± 0.0002	0.7702 ± 0.0003
Logit margin M (s)	0.3343 ± 0.0005	0.4913 ± 0.0001	0.7599 ± 0.0001
Subset selection (s)	0.0383 ± 0.0001	0.0086 ± 0.0004	0.0128 ± 0.0001
Stage subtotal (s)	0.7164 ± 0.0008	1.0016 ± 0.0005	1.5429 ± 0.0004
Peak scoring memory (MB)	2188.6	1067.8	2284.6
Full training of one surrogate			
Mean training time (s)	43.23	104.22	93.43
Mean peak GPU memory (GiB)	0.892	0.838	1.034

L Qualitative Evaluation of Attack Initialization

To qualitatively assess the selected base samples, we visualize three different test targets alongside the training images chosen for perturbation. Figure 5 shows each target paired with samples obtained through random selection and BASIS. The top rows show randomly selected samples, while the bottom rows show those selected by BASIS for the same targets. The examples reveal greater visual similarity between the BASIS-selected samples and their corresponding targets. The representation-space similarities are consistent with this pattern. Across the three targets, BASIS increases the average target–sample similarity from 0.267 to 0.359, 0.235 to 0.342, and 0.257 to 0.377, respectively.



(a) Target 3725.



(b) Target 7488.



(c) Target 3875.

Figure 5: Qualitative evaluation of base selection for three distinct test targets. Each panel displays the target image on the left. The corresponding selected samples are shown on the right with RAND on the top row and BASIS on the bottom. Numbers below each sample report its representation-space similarity to the target, with the average similarity shown on the right.